PLANEO DE UN AVIÓN

Aproximaciones a la solución exacta

Bloque I - Física Computacional

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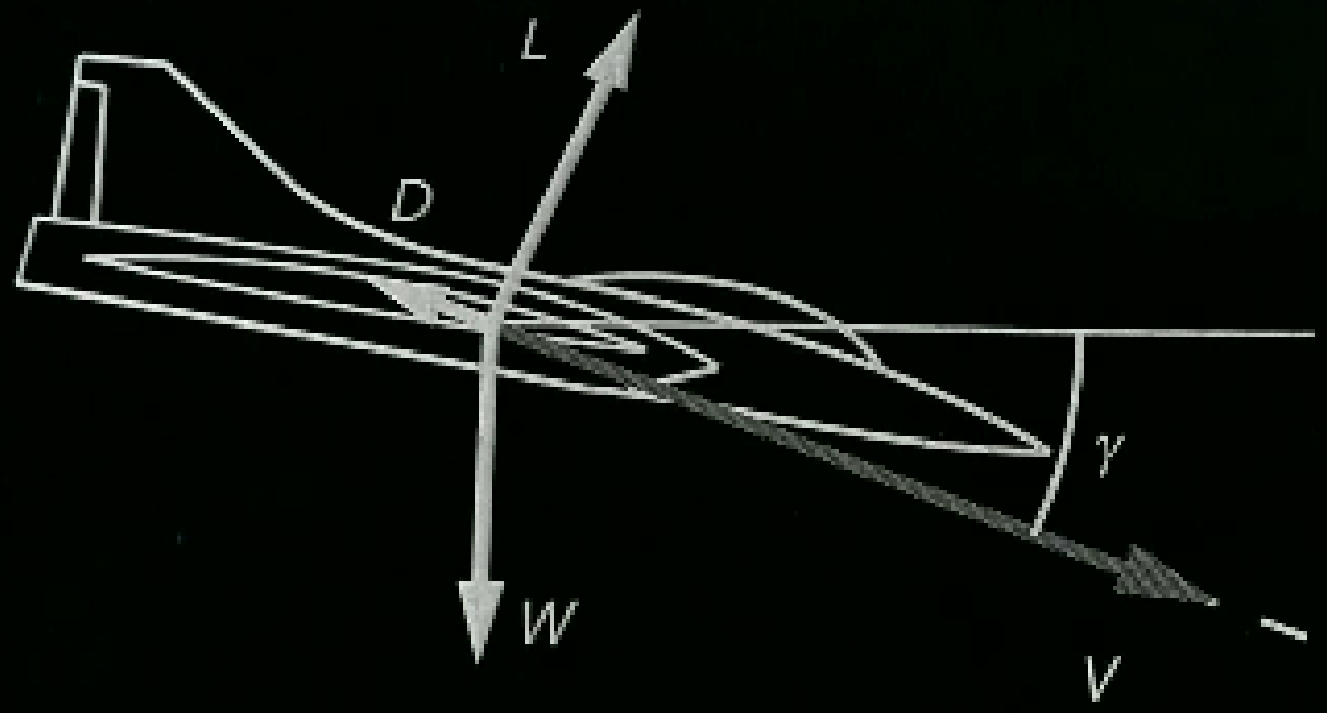
21 de diciembre del 2023

¿QUÉ VAMOS A HACER?

Problema a resolver - Ecuaciones

2da Ley de Newton

$$\sum_k \vec{F}_k = m\vec{a} = m\dot{\vec{v}} = \frac{d\vec{p}}{dt}$$


L
D
γ
W
V

Problema simplificado

$$m\dot{v} = T - D - gmSen\left(\gamma\right)$$
$$m\dot{\gamma} = \frac{L}{v} - \frac{g}{v}mCos\left(\gamma\right)$$
$$\dot{h} = vSen\left(\gamma\right)$$
$$\dot{x} = vCos\left(\gamma\right)$$

Problema simplificado

Donde

$$D = 0.5 \cdot \rho \cdot v^2 \cdot S \cdot CD$$
$$L = 0.5 \cdot \rho \cdot v^2 \cdot S \cdot CL$$
$$T = 0$$

Problema simplificado

$$\dot{v} = -\frac{0'5 \cdot \rho \cdot S \cdot CD}{m} v^2 - gSen\left(\gamma\right)$$

$$\dot{\gamma} = \frac{0'5 \cdot \rho \cdot S \cdot CL}{m} v - \frac{g}{v} Cos\left(\gamma\right)$$

$$\dot{h} = vSen\left(\gamma\right)$$

$$\dot{x} = vCos\left(\gamma\right)$$

SOLUCIÓN ITERANDO

Problema a resolver

Ecuaciones iteradas

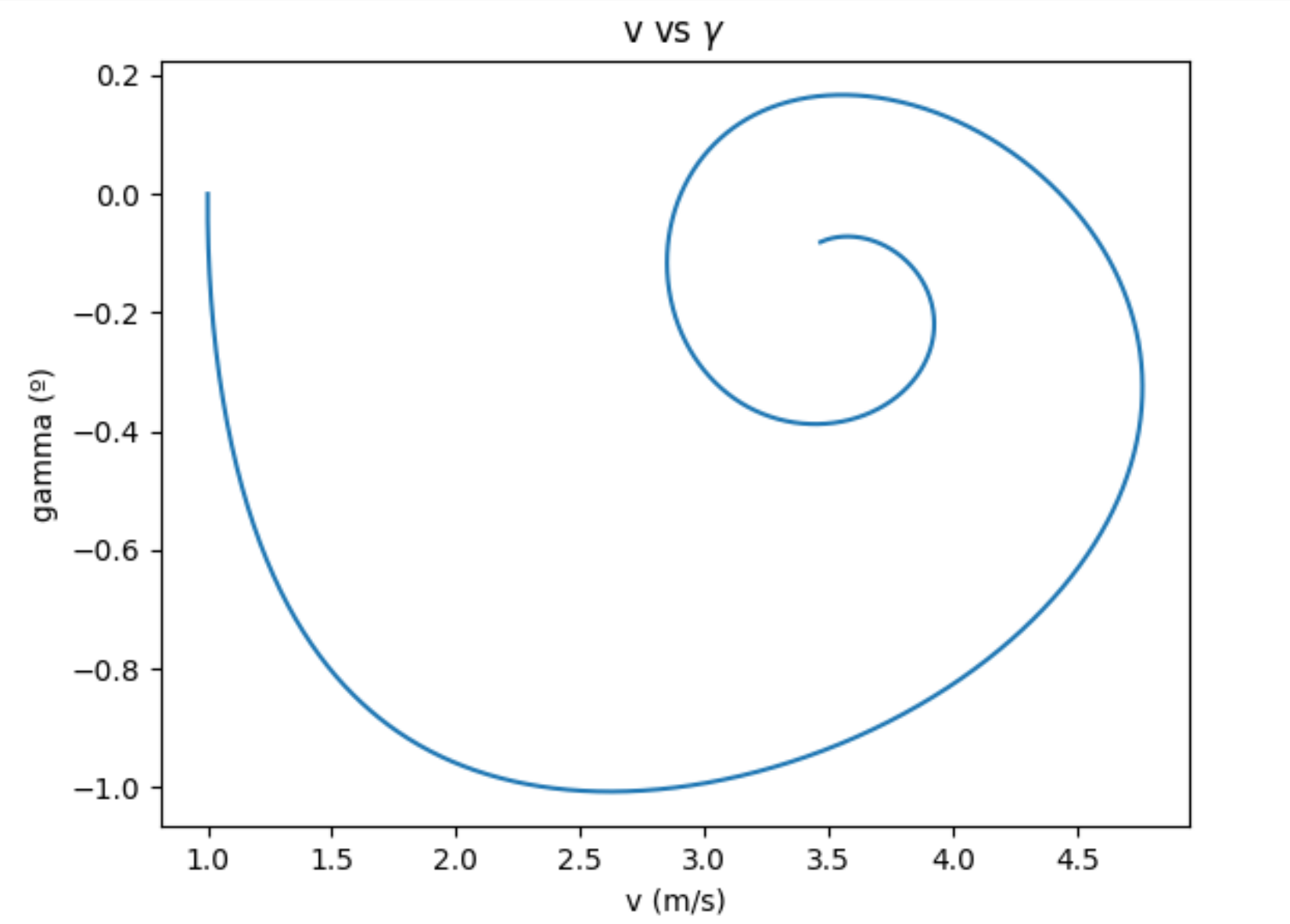
$$v^{k+1} = v^k + \Delta t \left(-\frac{0'5 \cdot \rho \cdot S \cdot CD}{m} (v^k)^2 - gSen\gamma^k \right)$$

$$\gamma^{k+1} = \gamma^k + \Delta t \left(\frac{0'5 \cdot \rho \cdot S \cdot CL}{m} v^k - \frac{g}{v} Cos\left(\gamma^k\right) \right)$$

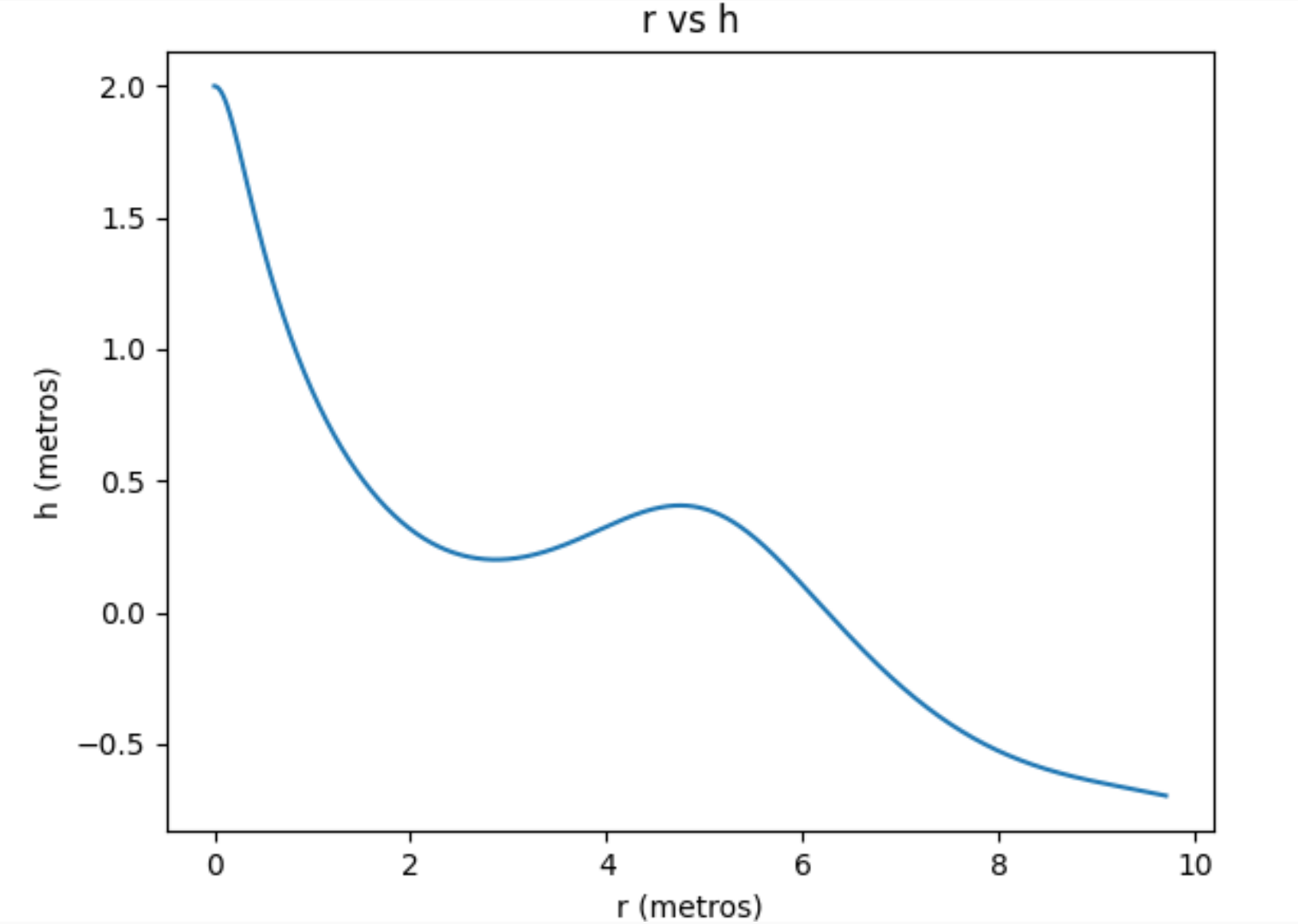
$$h^{k+1} = h^k + \Delta t \cdot v^k \cdot Sen\gamma^k$$

$$x^{k+1} = x^k + \Delta t \cdot v^k \cdot Cos\gamma^k$$

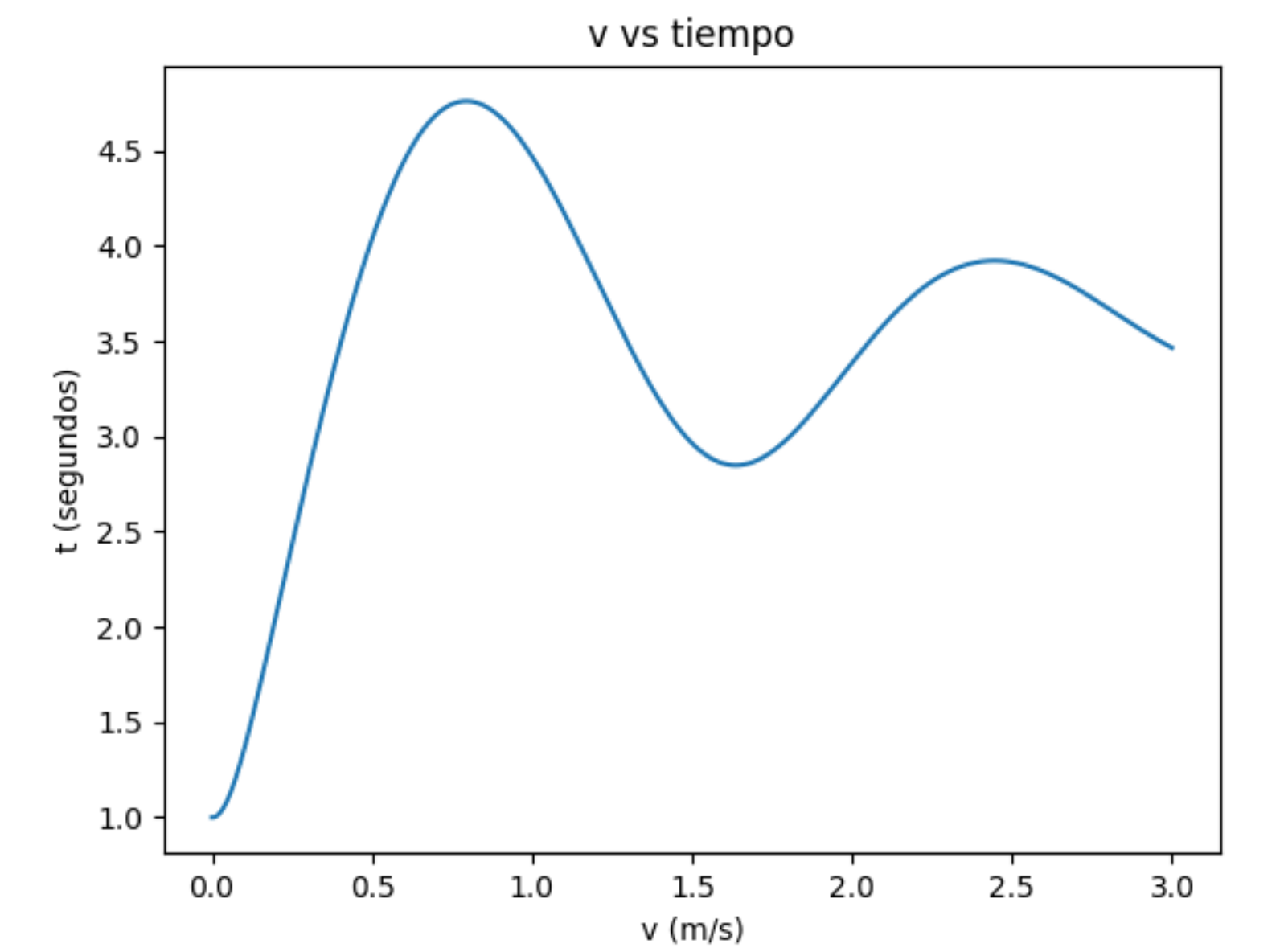
Ángulo frente a velocidad



Trayectoria

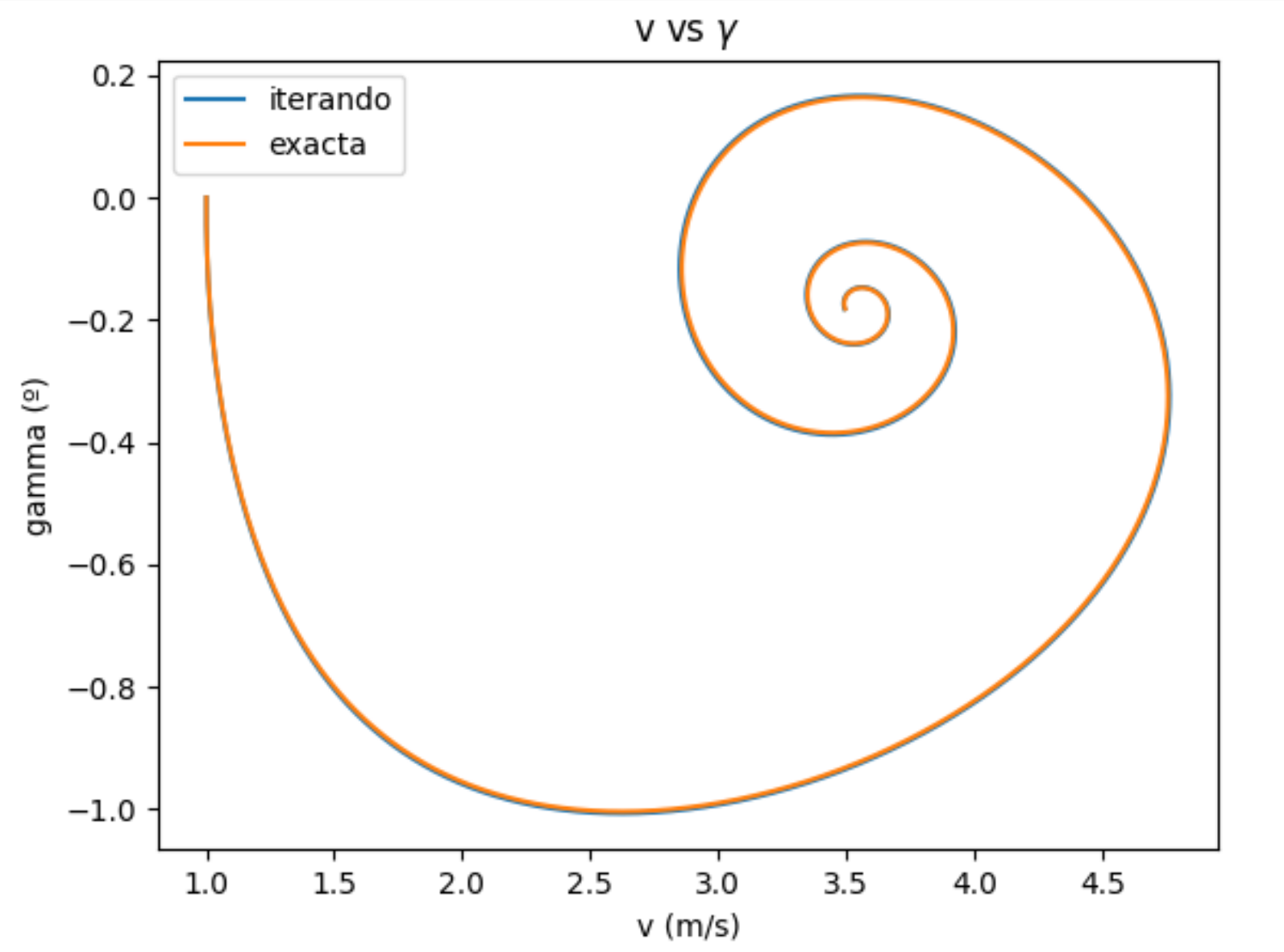


Velocidad frente al tiempo

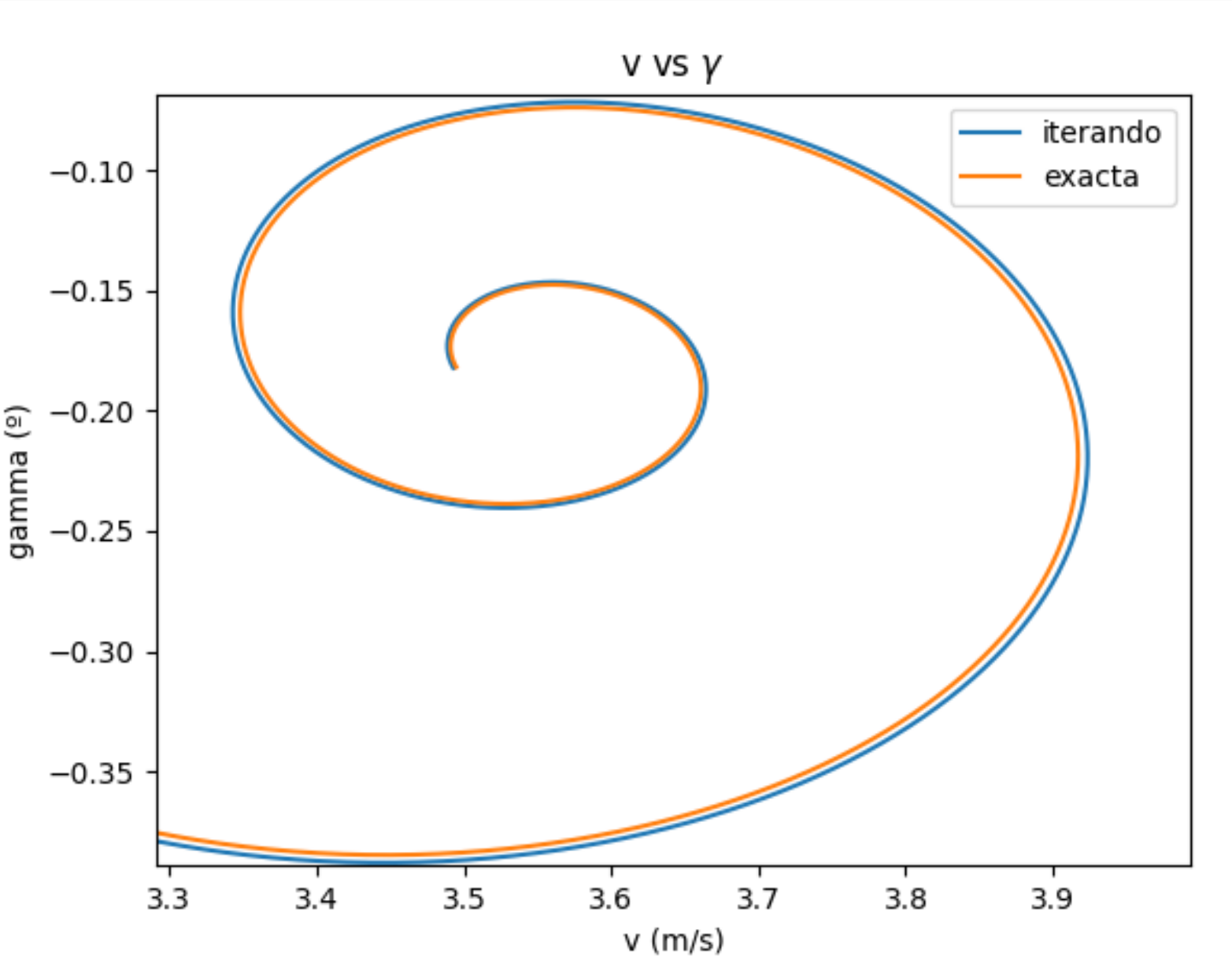


COMPARANDO CON EXACTA

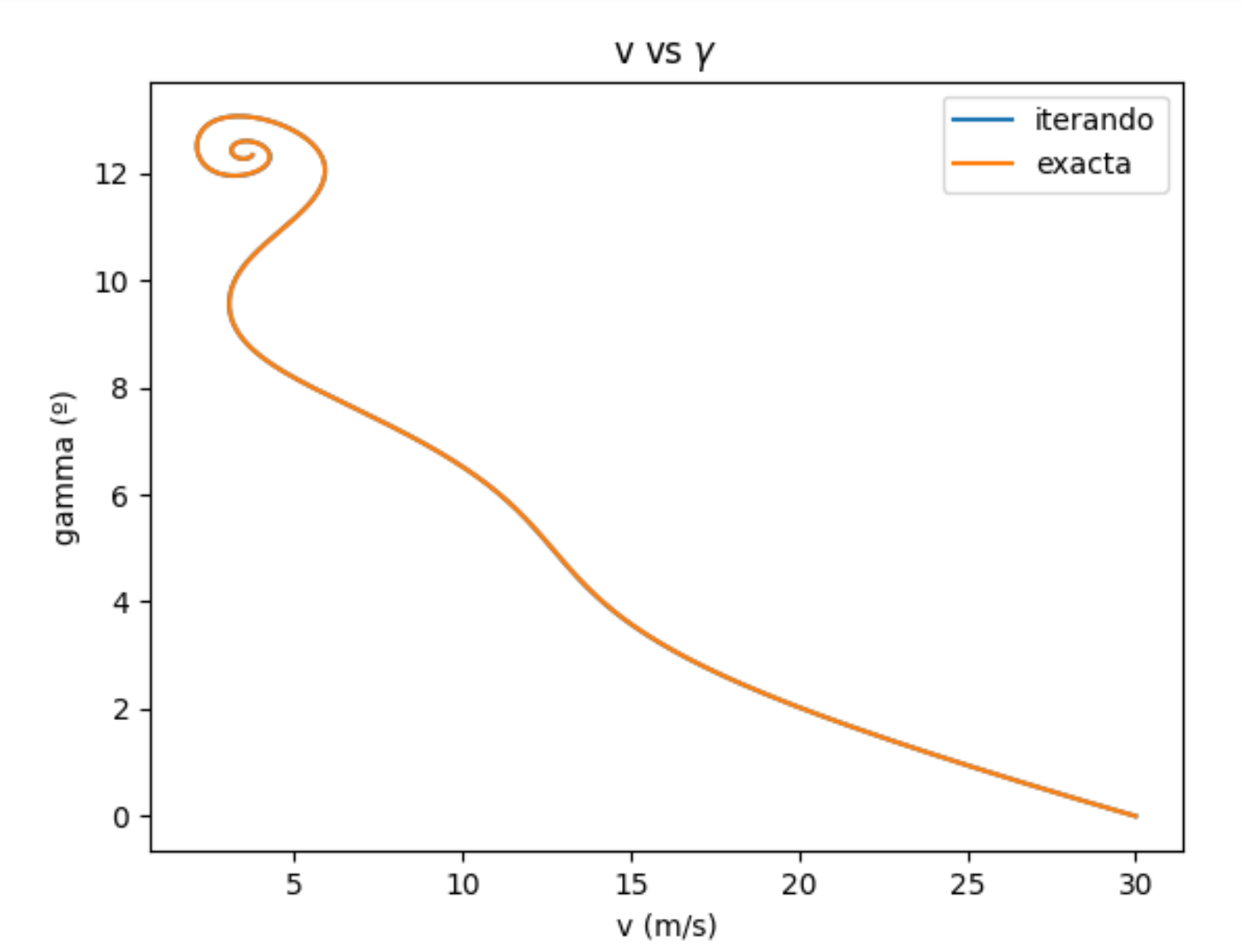
Ángulo frente a velocidad



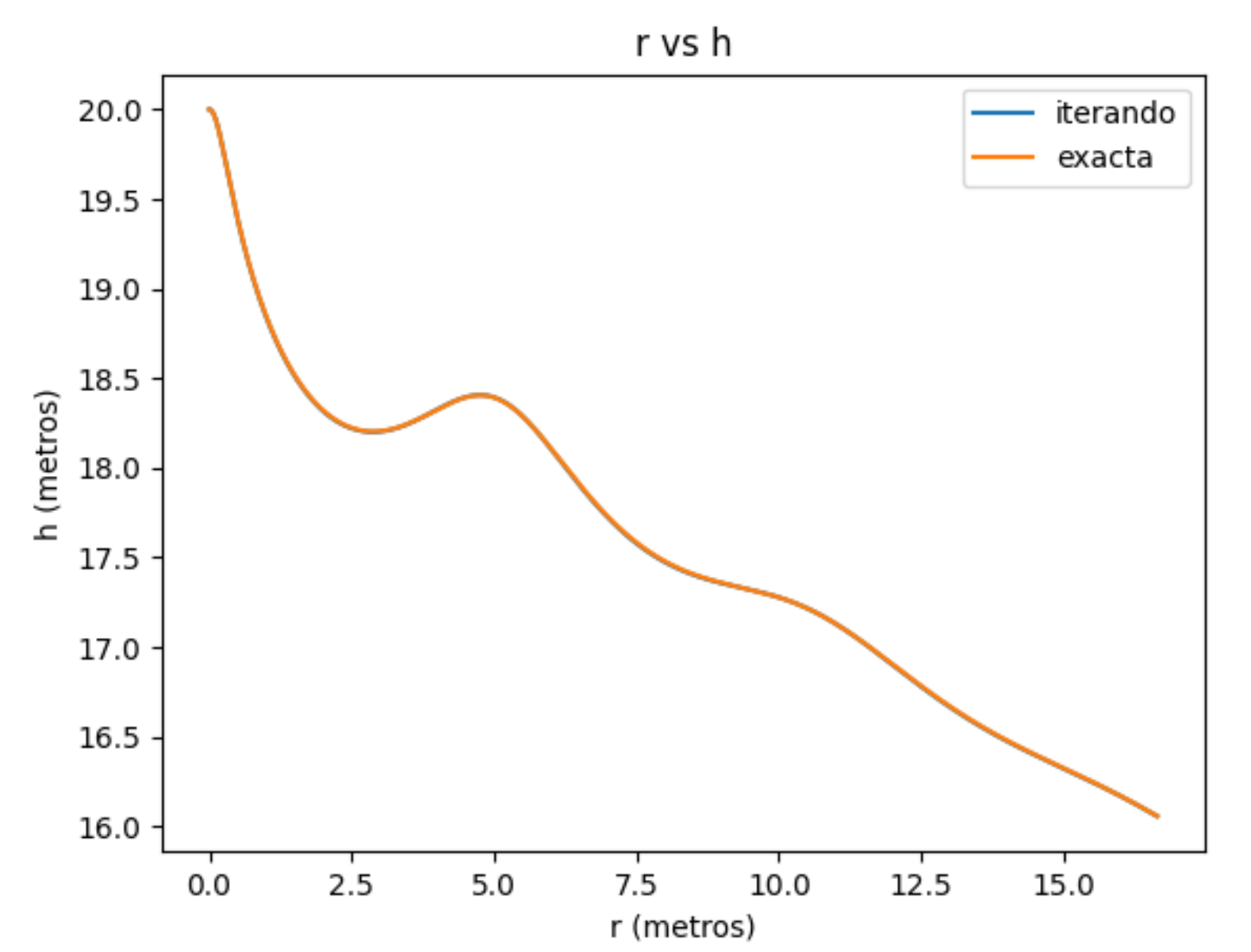
Ángulo frente a velocidad



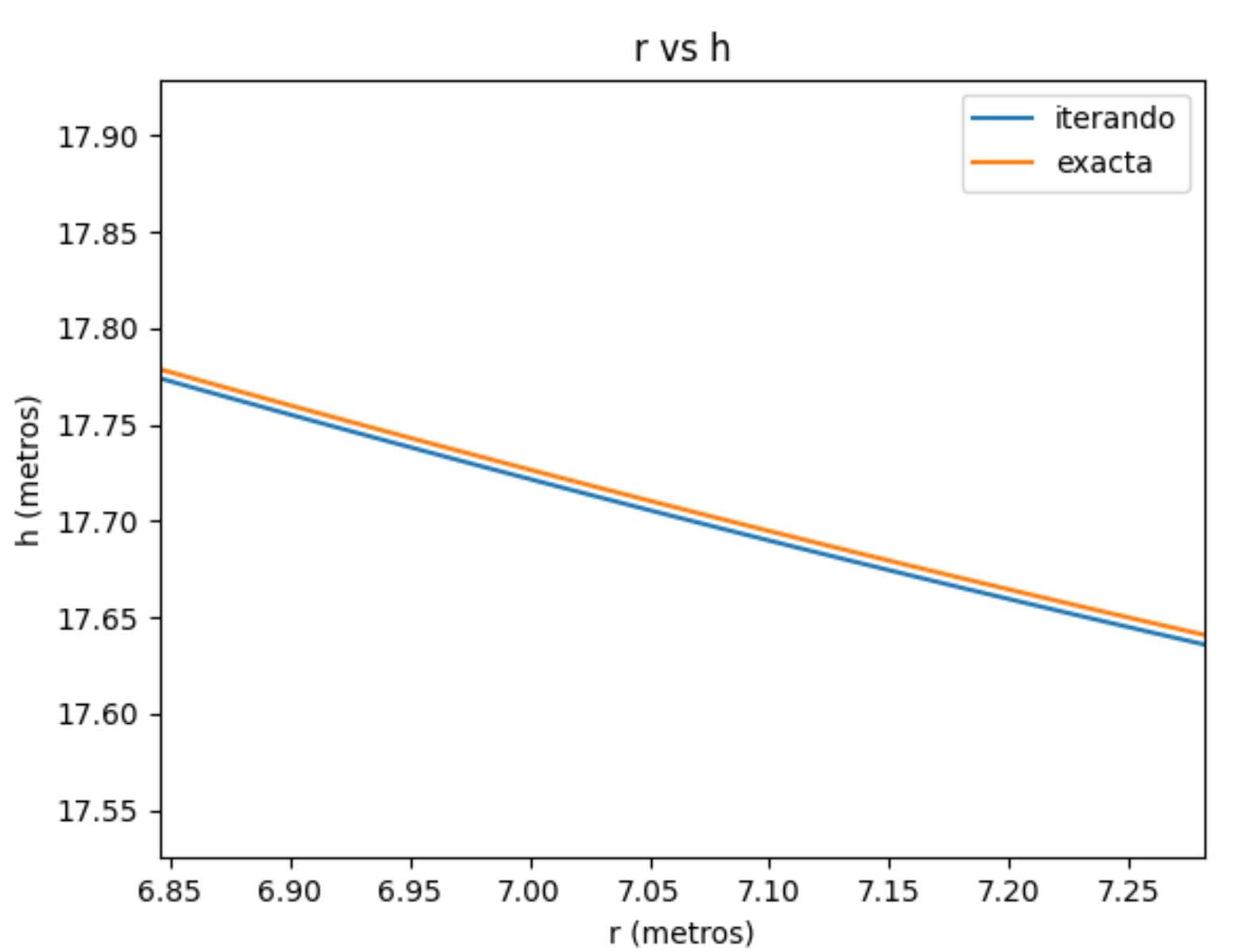
Ángulo frente a velocidad, v = 30 m/s



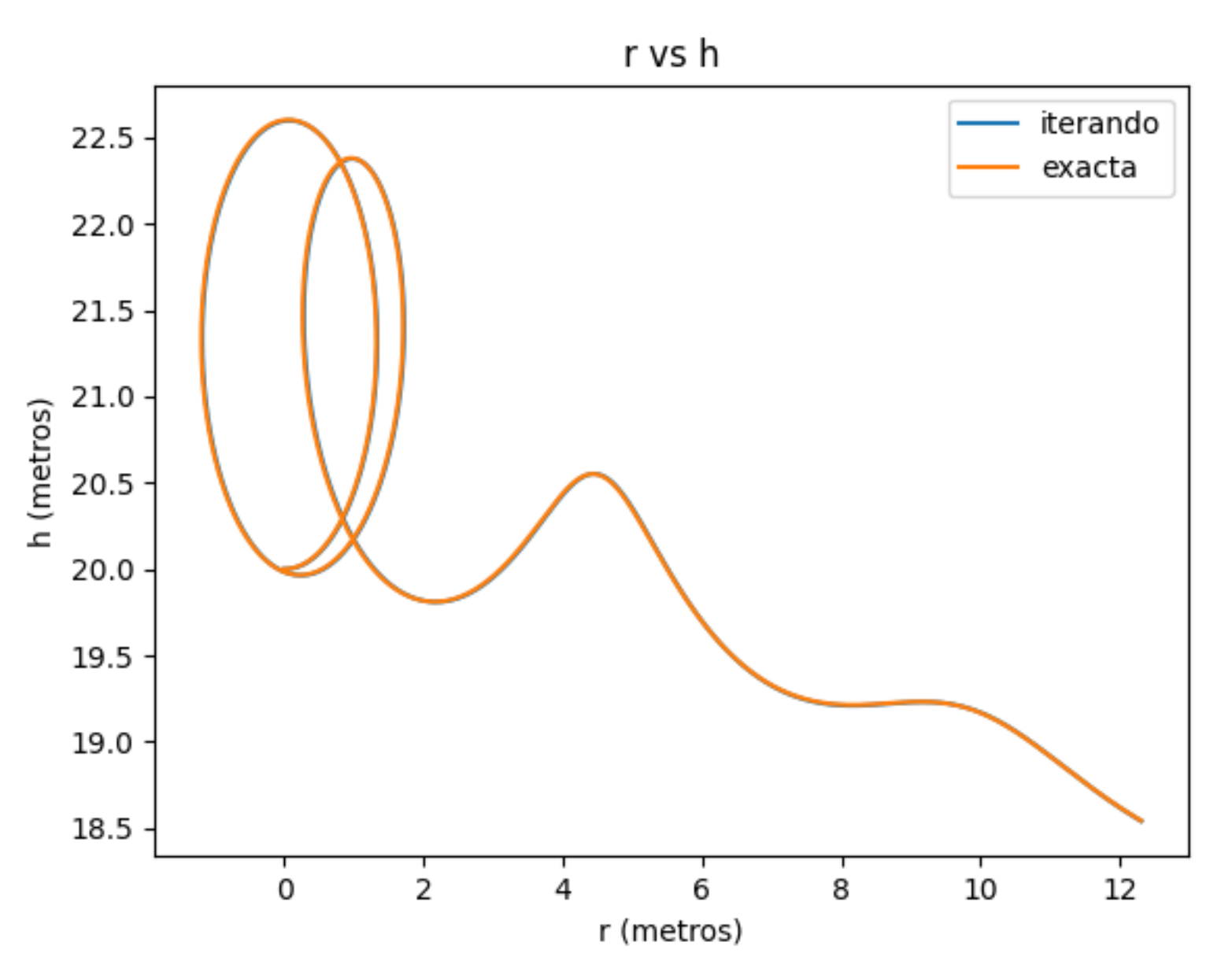
Trayectoria



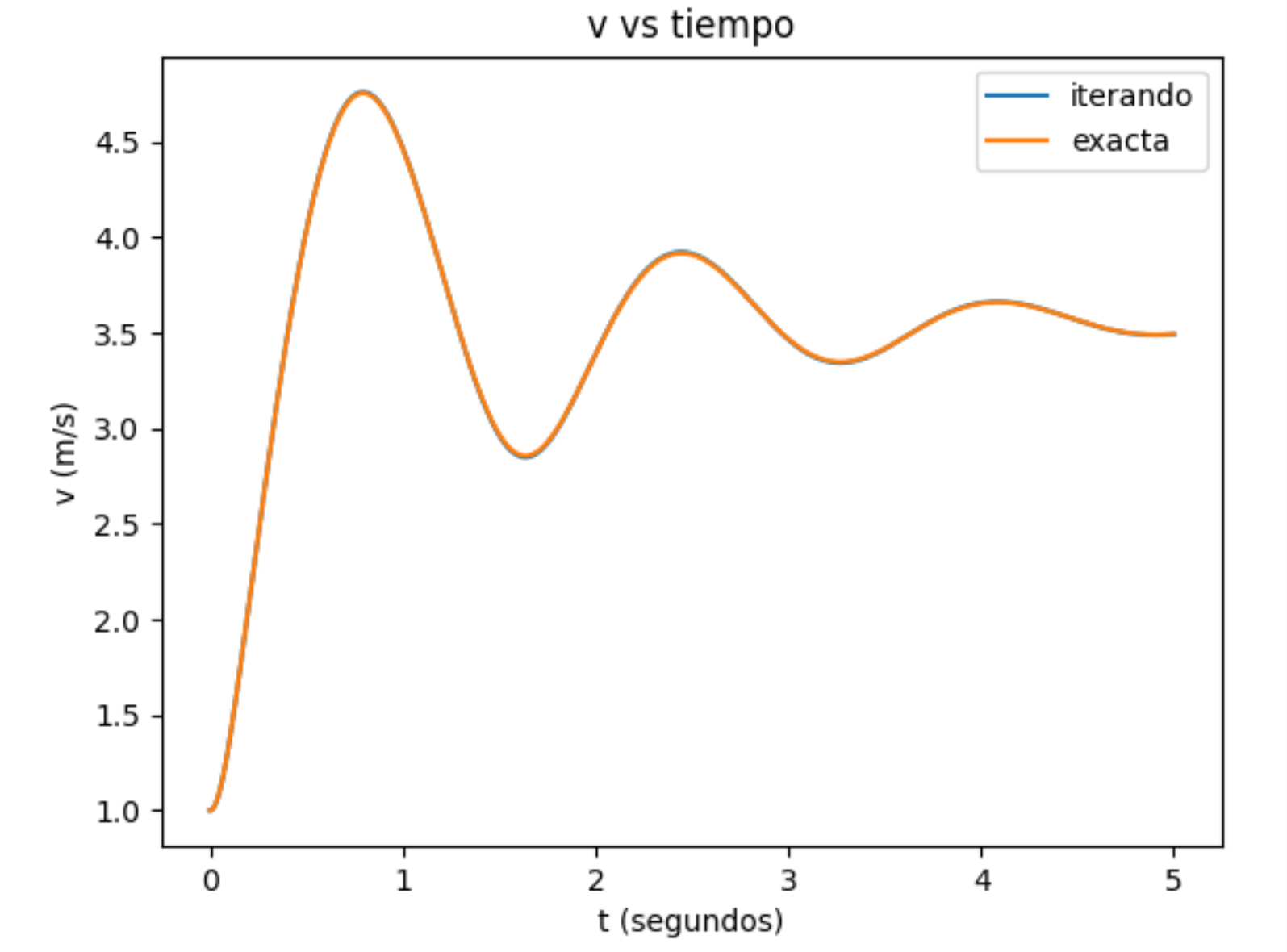
Trayectoria



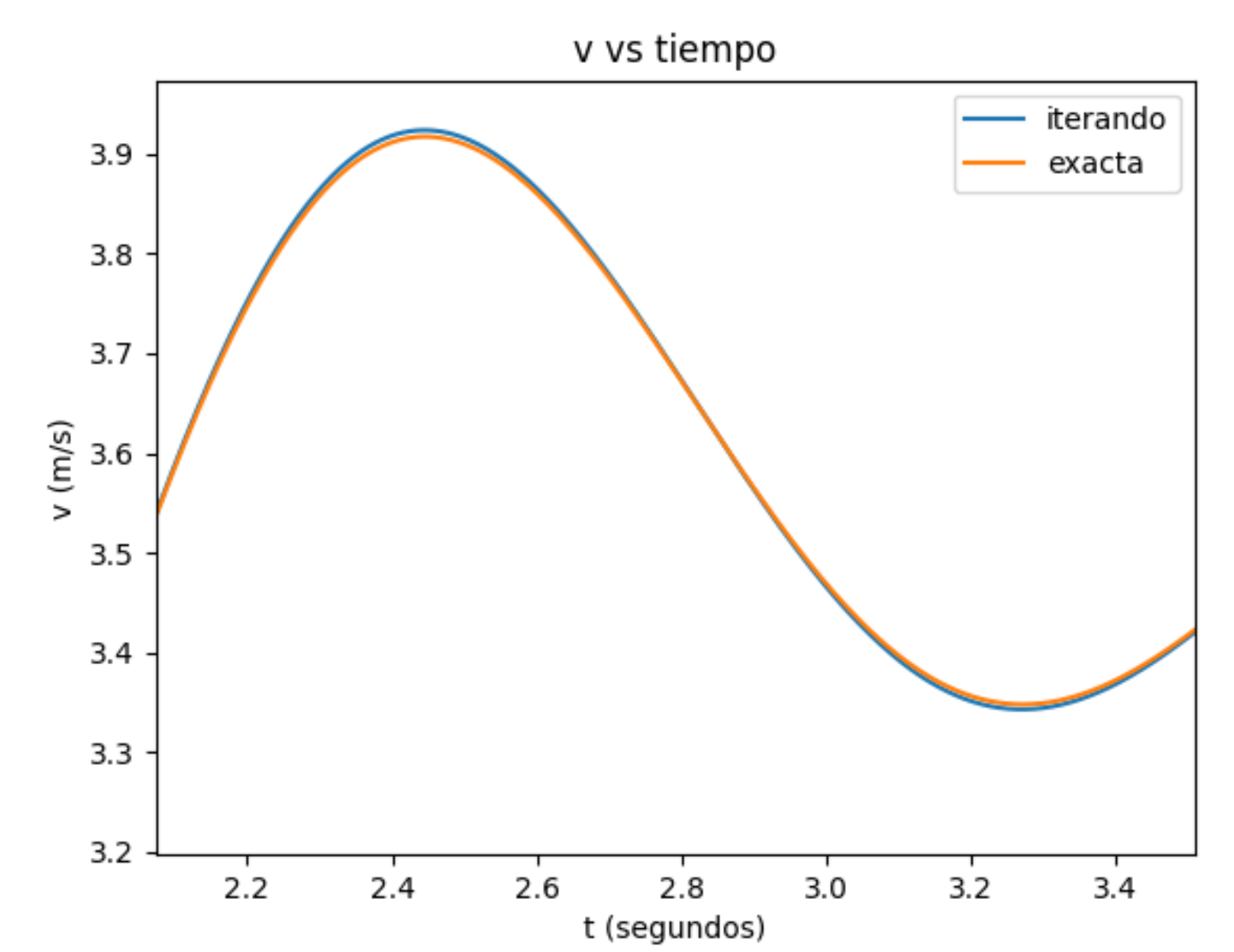
Trayectoria, v = 30 m/s



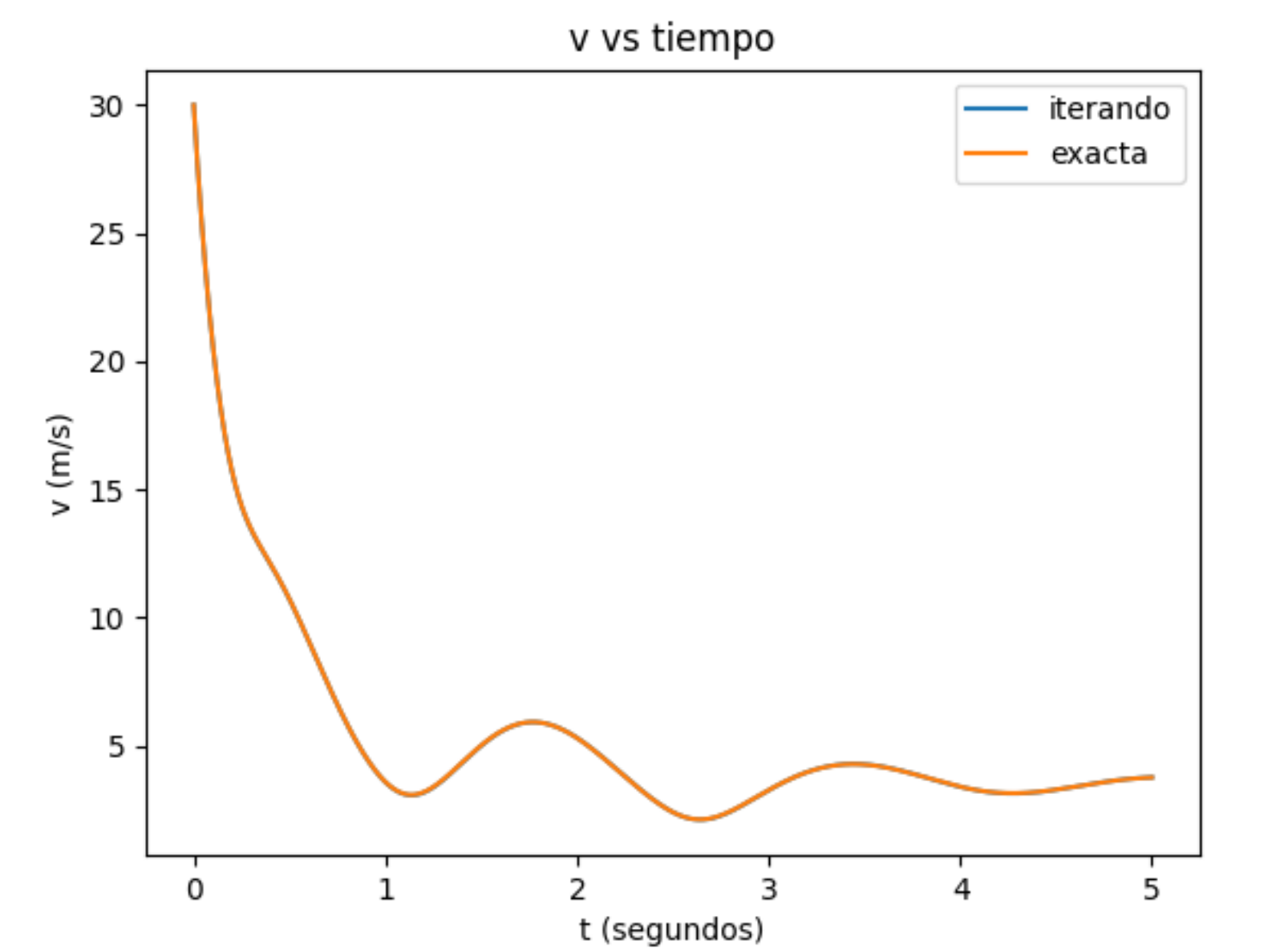
Velocidad frente al tiempo



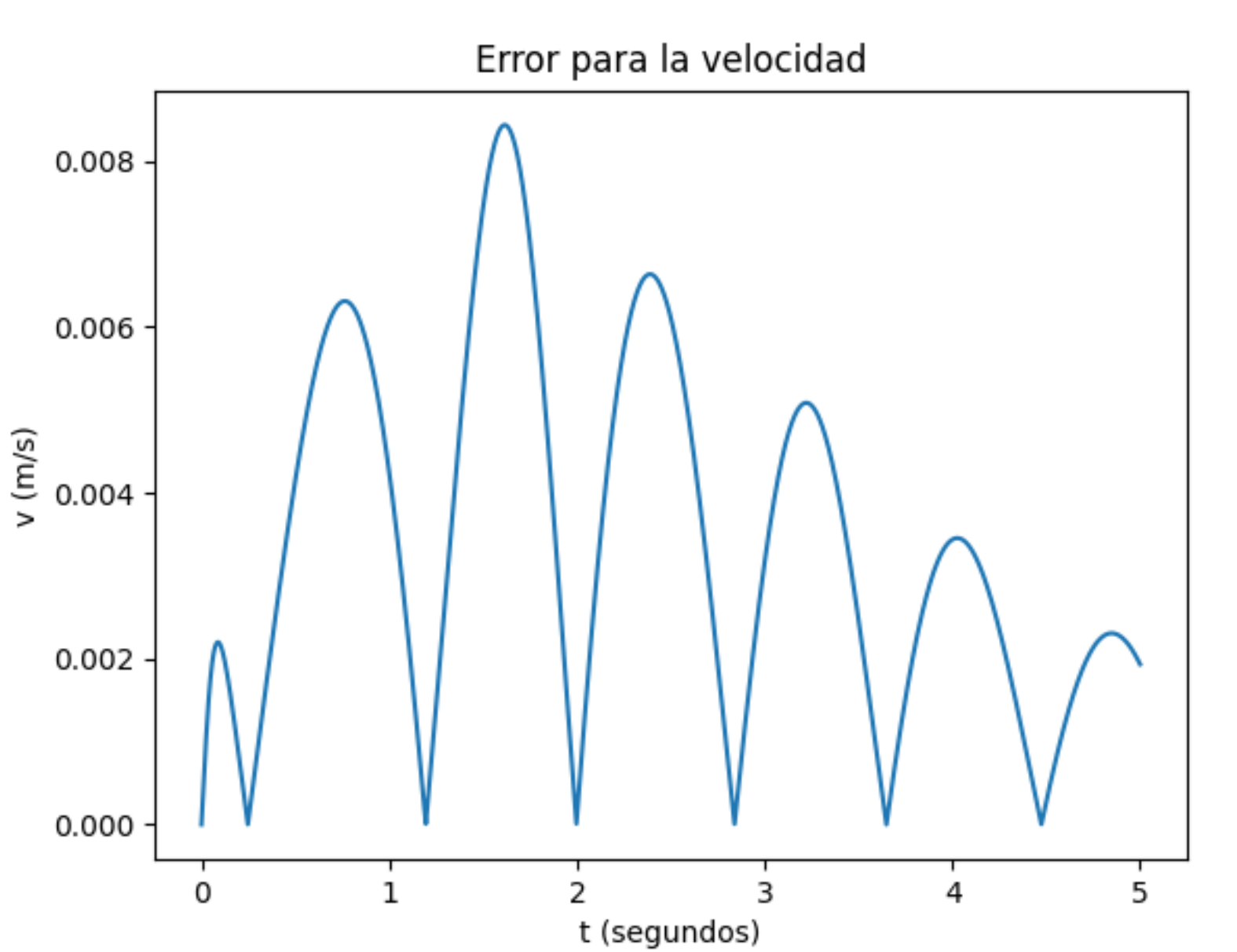
Velocidad frente al tiempo



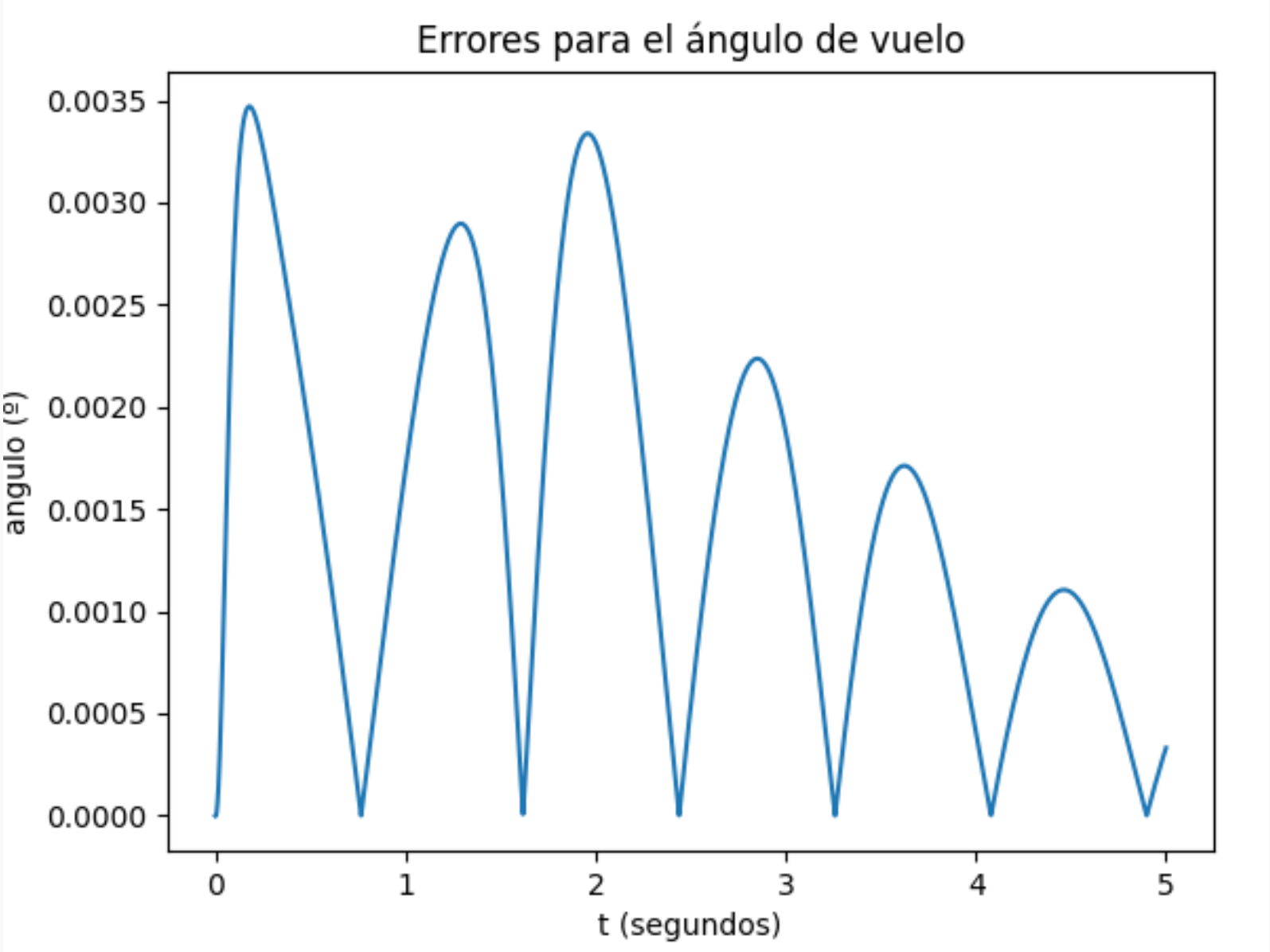
Velocidad frente al tiempo,v = 30m/s



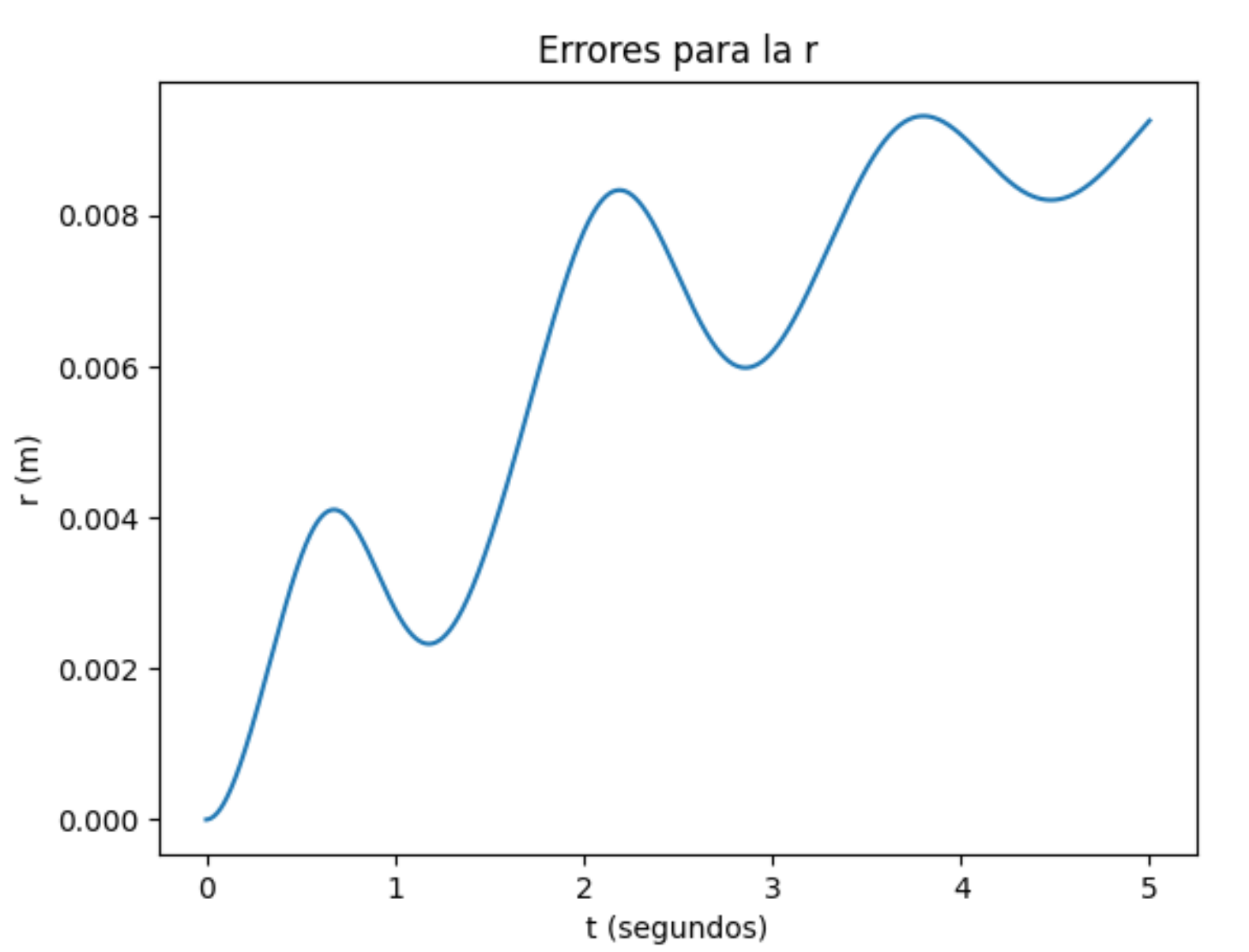
Error para la velocidad



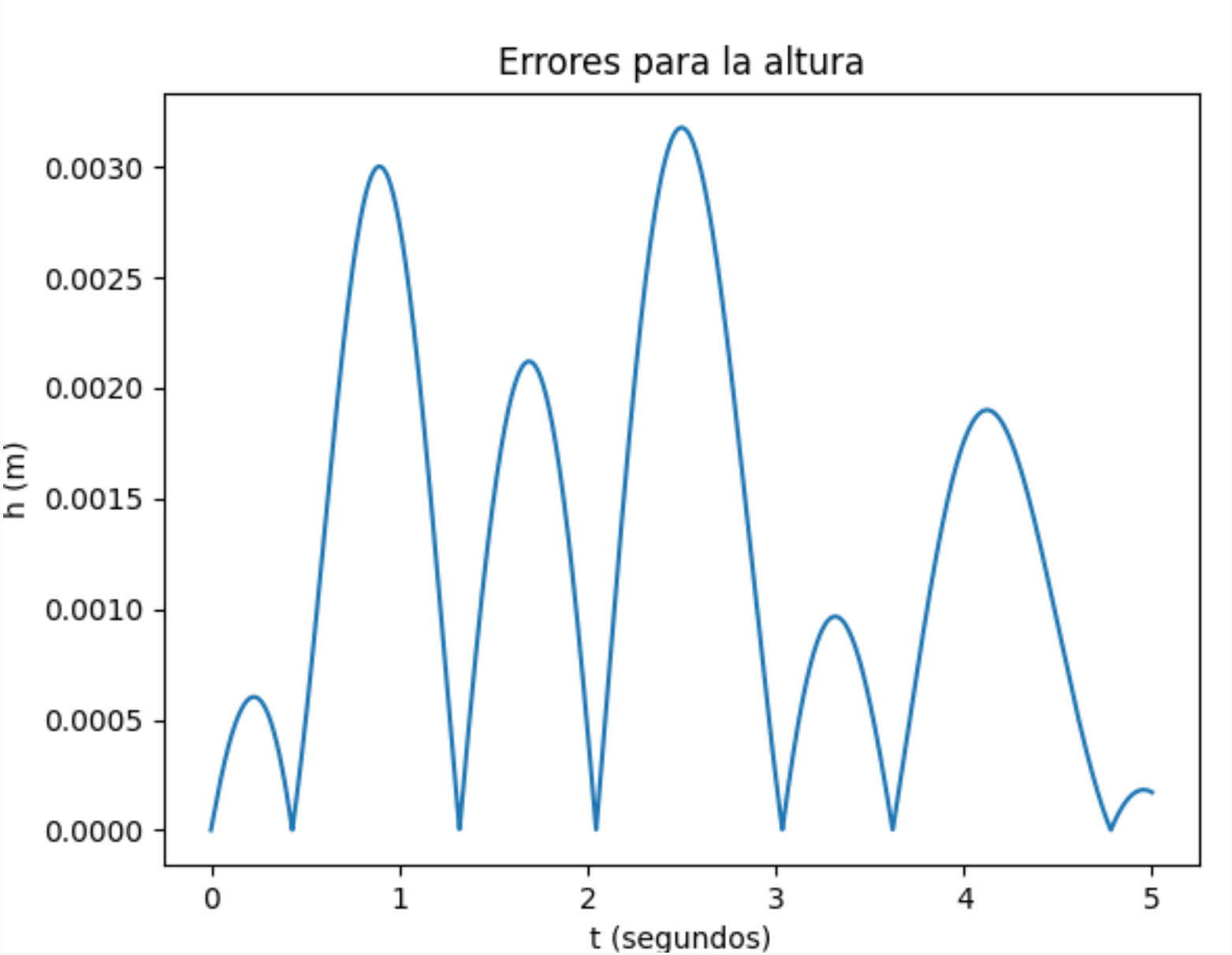
Error para el ángulo



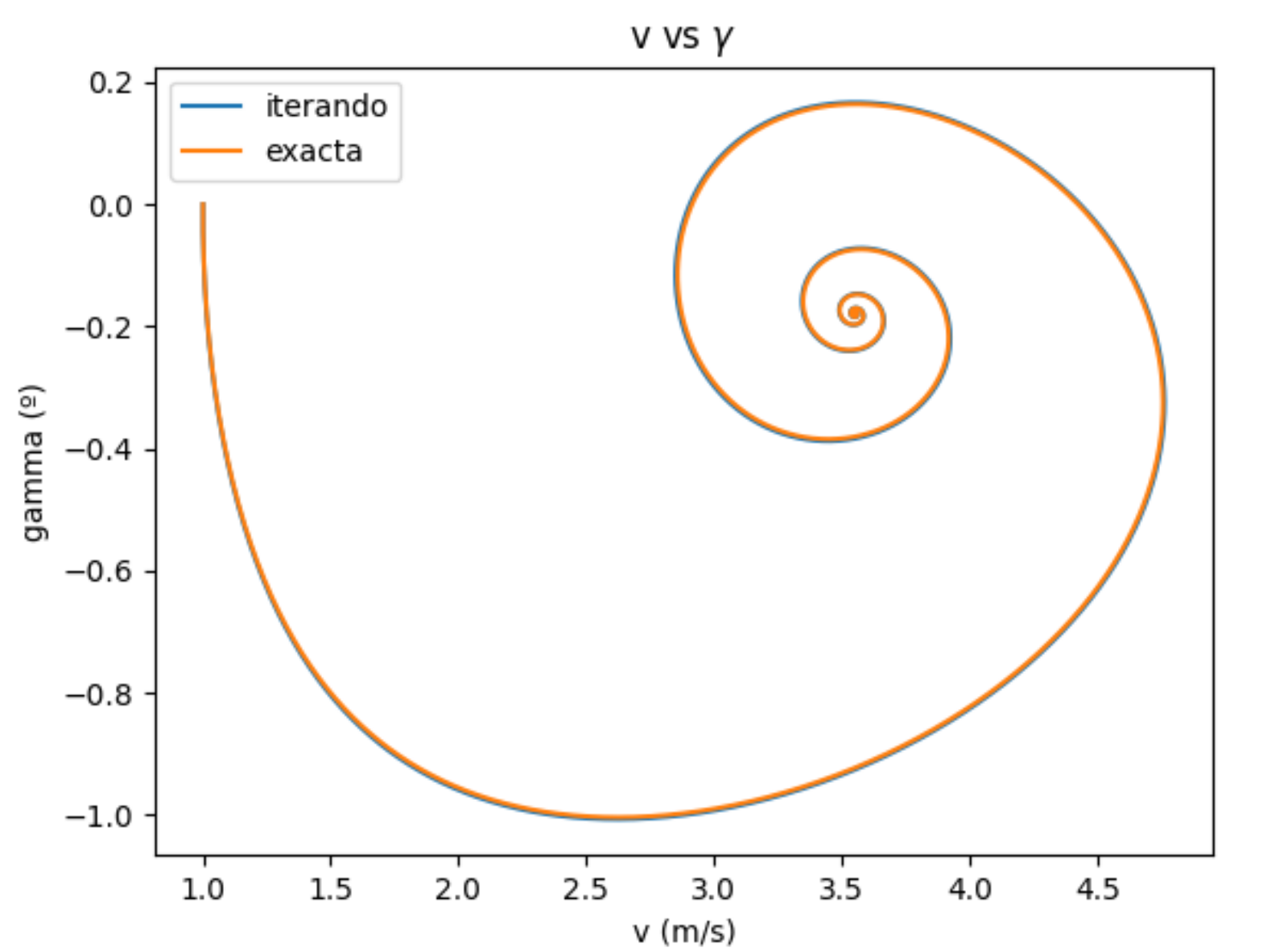
Error para la r



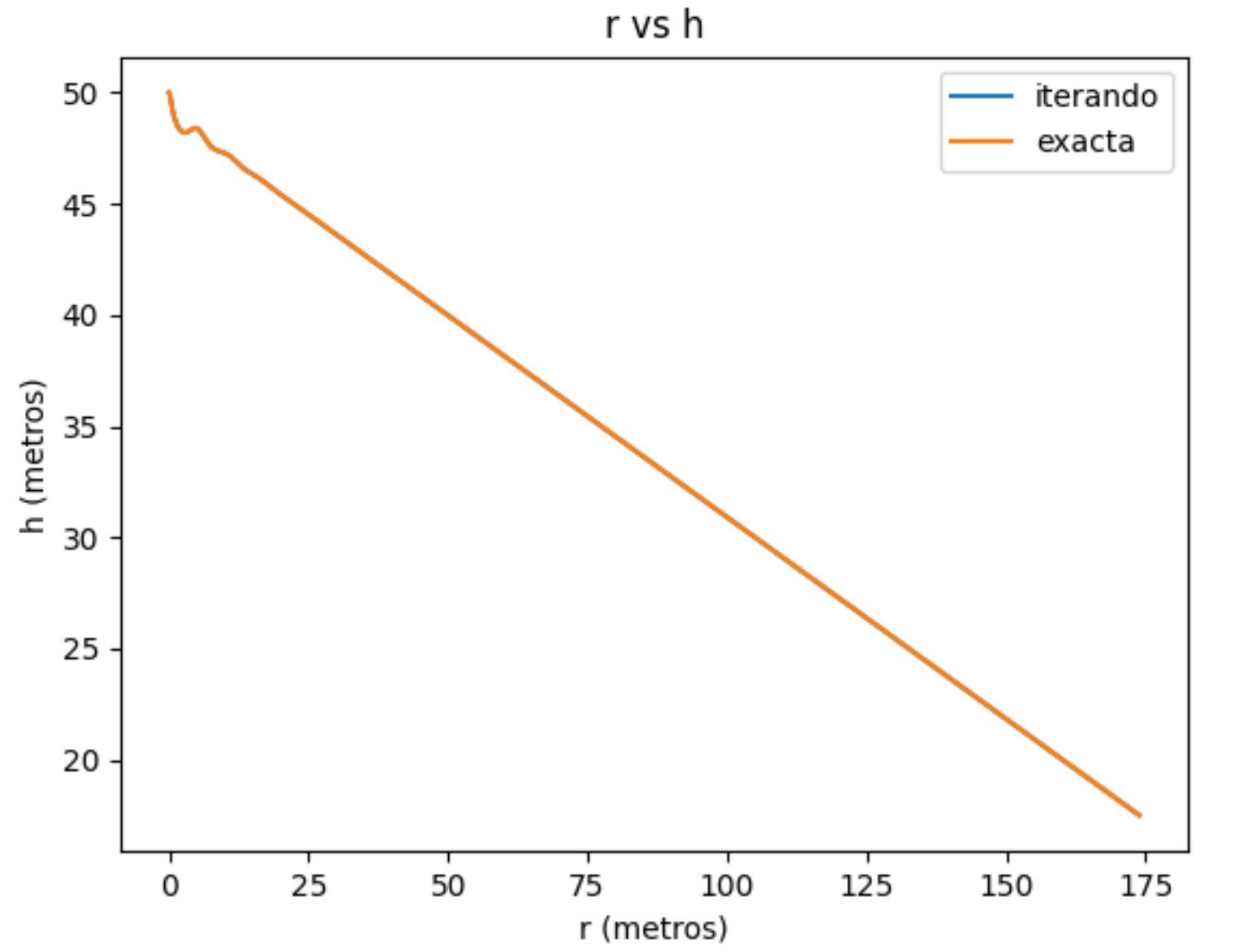
Error para la altura



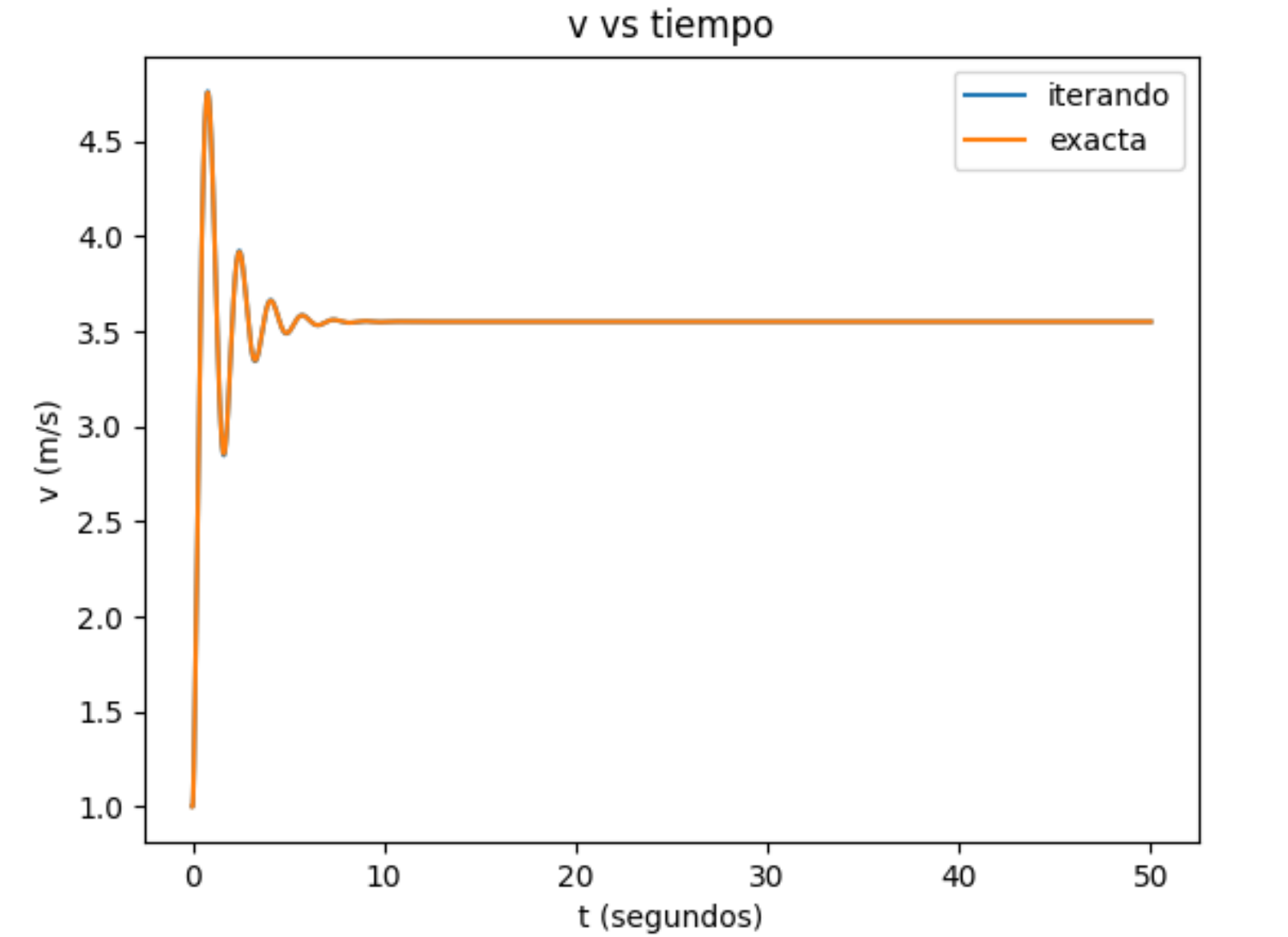
Comportamiento tiempos grandes



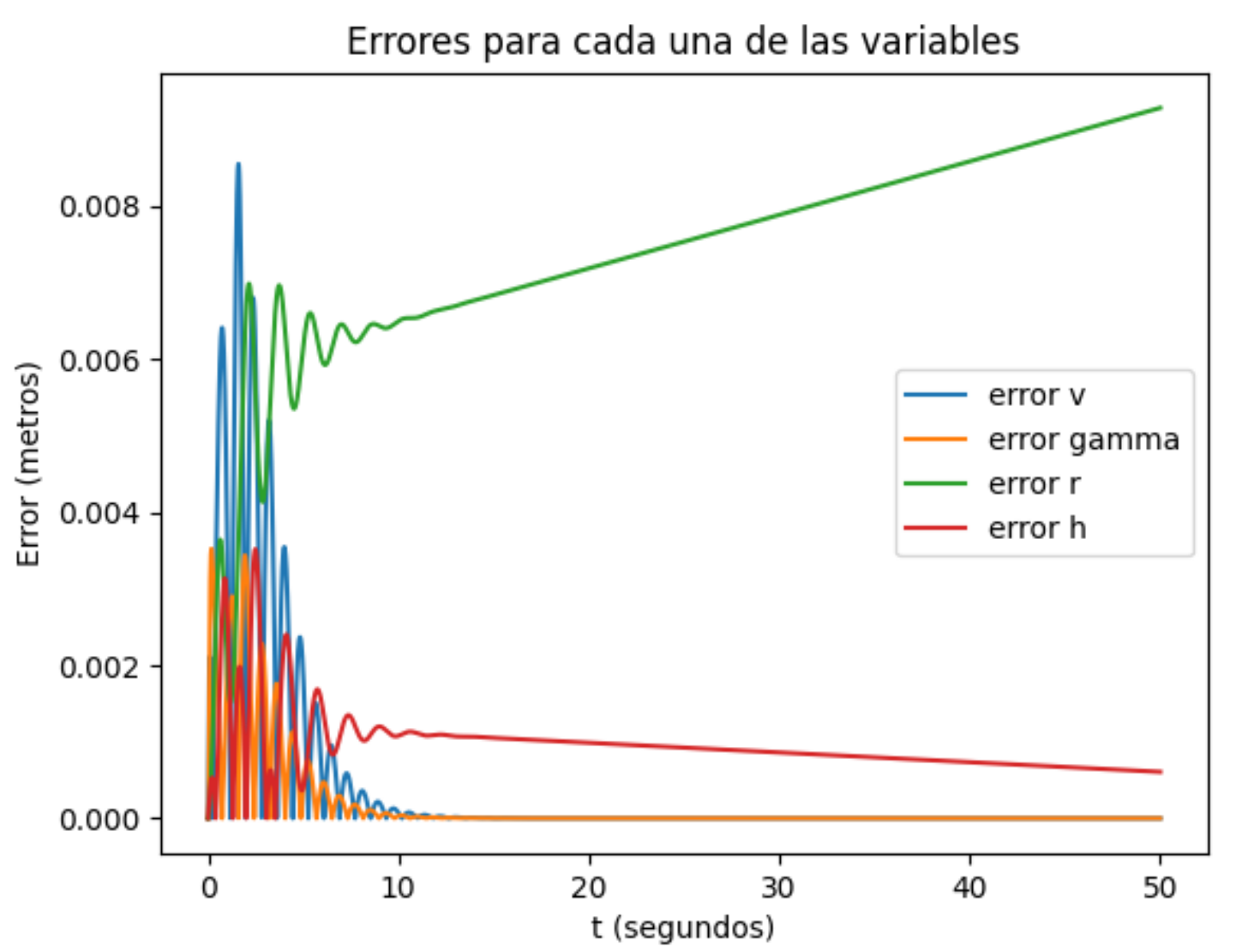
Comportamiento tiempos grandes



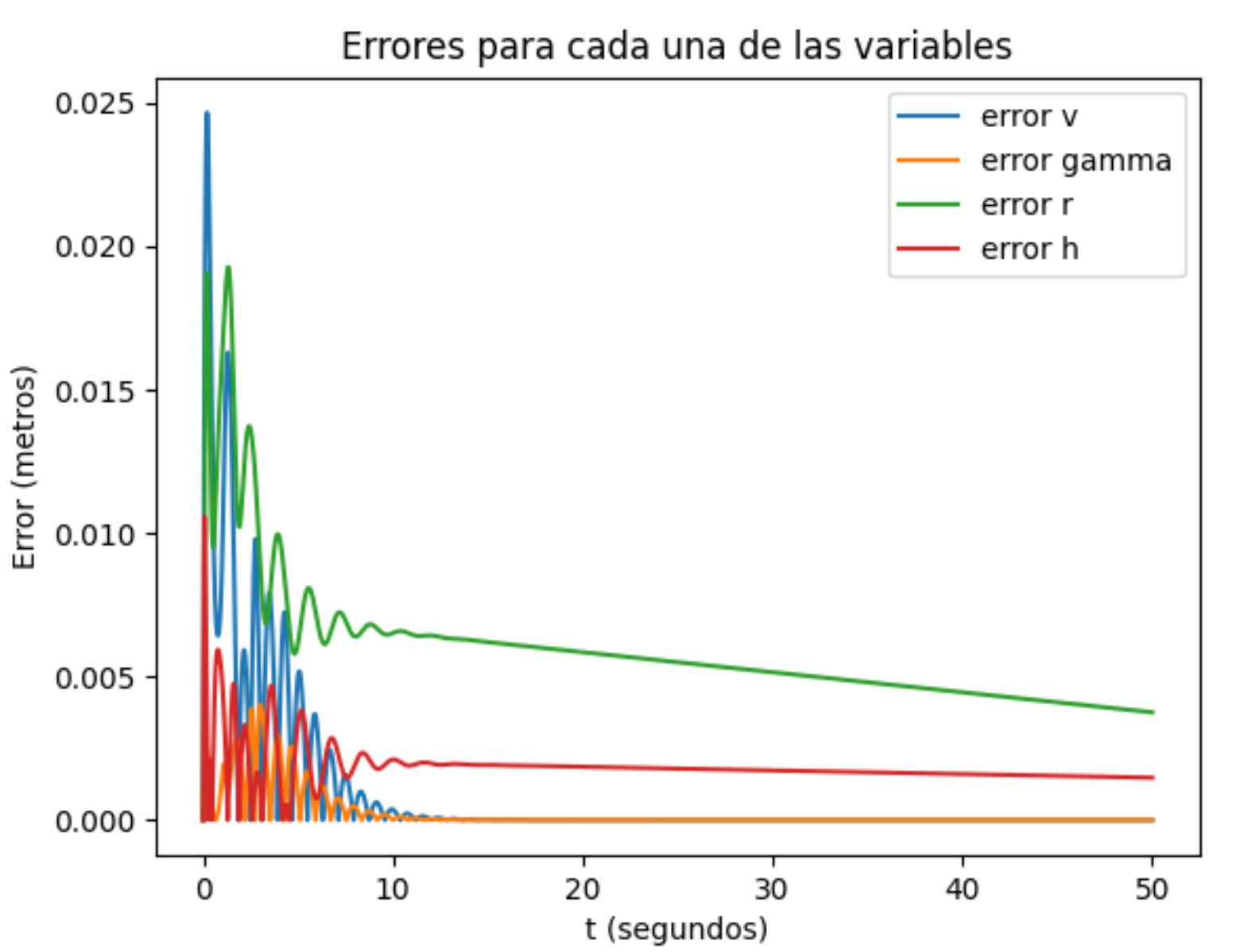
Comportamiento tiempos grandes



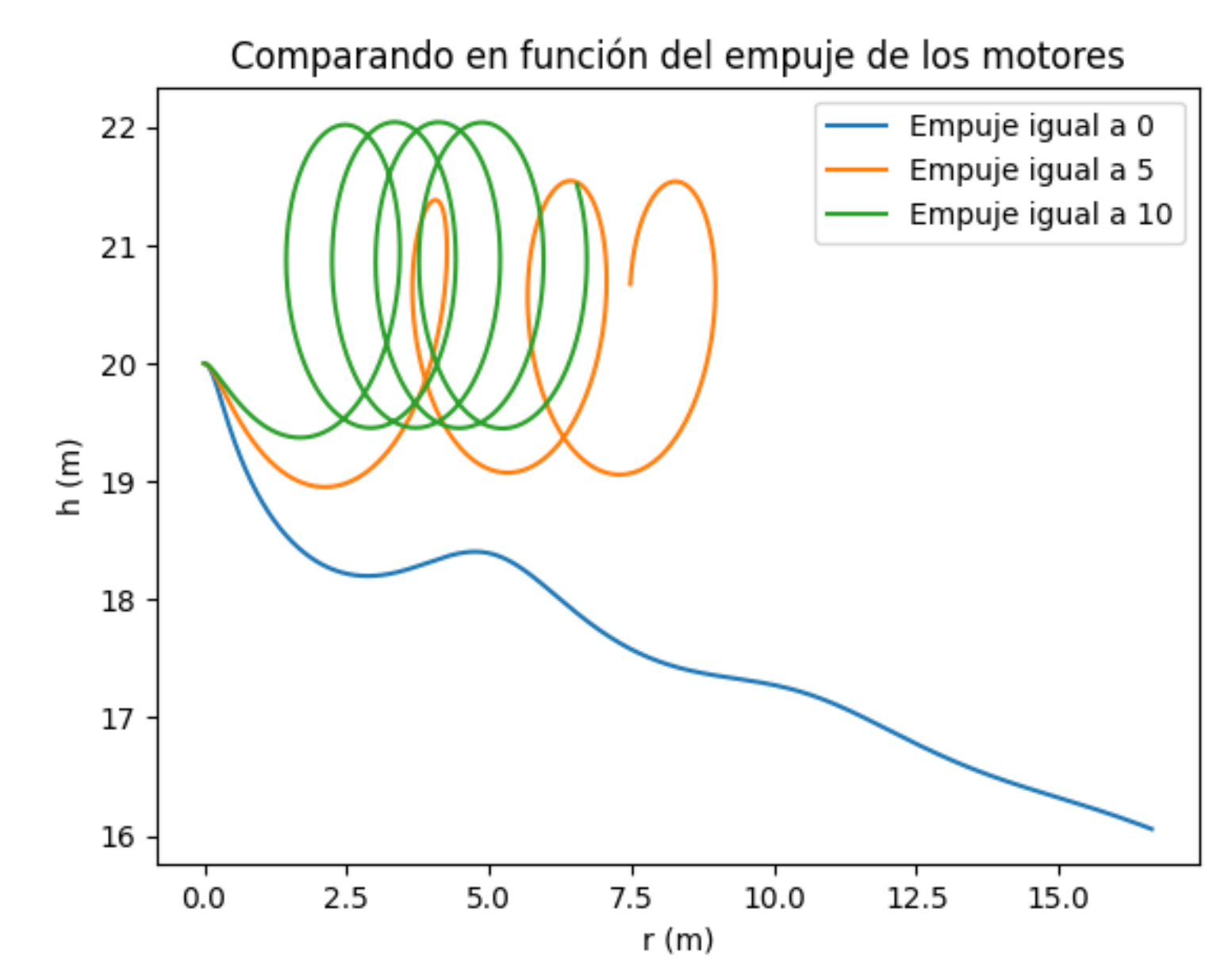
Comportamiento tiempos grandes



Tiempos grandes, v = 30 m/s



Algunas trayectorias con motores



APROXIMACIÓN PRIMER ORDEN

Ecuaciones a primer orden

$$\dot{v} = -\frac{0'5 \cdot \rho \cdot S \cdot CD}{m} v^2 - gSen\left(\gamma\right)$$

$$\dot{\gamma} = \frac{0'5 \cdot \rho \cdot S \cdot CL}{m} v - \frac{g}{v} Cos\left(\gamma\right)$$

$$\dot{h} = vSen\left(\gamma\right)$$

$$\dot{x} = vCos\left(\gamma\right)$$

$$\frac{d}{dt}u = Au + B$$

u = [v, γ, h, x]

Desarrollos a primer orden

$$v^2 =_{v_0} v_0^2 + 2v_0(v - v_0) + O^2(v) \approx_{v_0} 2v_0 v - v_0^2$$

$$Sen\gamma =_0 Sen0 + \gamma Cos0 + O^2(\gamma) \approx_0 \gamma$$

$$\frac{Cos\gamma}{v} =_{(v_0,0)} \frac{1}{v_0} - \gamma\frac{Sen0}{v_0} - \frac{v-v_0}{v_0^2}Cos0 + O^2(v,\gamma) \approx_{(v_0,0)}$$

$$\approx_{(v_0,0)} \frac{2}{v_0} - \frac{v}{v_0^2}$$

$$vSen\gamma =_{(v_0,0)} v_0 Sen0 + (v - v_0)Sen0 + v\gamma Cos0 +$$

$$+O^2(v,\gamma) \approx_{(v_0,0)} v_0\gamma$$

$$vCos\gamma =_{(v_0,0)} v_0 + (v - v_0)Cos0 - v_0\gamma Sen0 +$$

$$+O^2(v,\gamma) \approx_{(v_0,0)} v$$

Ecuaciones a primer orden

$$\begin{bmatrix} \dot{v} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \frac{-\rho \cdot v_0 \cdot S \cdot CD}{m} & -g \\ \frac{0.5 \cdot \rho \cdot S \cdot v_0^2 + m \cdot g}{m \cdot v_0^2} & 0 \end{bmatrix} \begin{bmatrix} v \\ \gamma \end{bmatrix} + \begin{bmatrix} \frac{\rho \cdot v_0 \cdot S \cdot CD}{m} \\ -2\frac{g}{v_0} \end{bmatrix}$$

Método explícito

$$\begin{bmatrix} \frac{v^{k+1}-v^k}{\Delta t} \\ \frac{\gamma^{k+1}-\gamma^k}{\Delta t} \end{bmatrix} = \begin{bmatrix} \frac{-\rho \cdot v_0 \cdot S \cdot CD}{m} & -g \\ \frac{0.5 \cdot \rho \cdot S \cdot v_0^2 + m \cdot g}{m \cdot v_0^2} & 0 \end{bmatrix} \begin{bmatrix} v^k \\ \gamma^k \end{bmatrix} + \begin{bmatrix} \frac{\rho \cdot v_0 \cdot S \cdot CD}{m} \\ -2\frac{g}{v_0} \end{bmatrix}$$

$$\begin{bmatrix} v^{k+1} \\ \gamma^{k+1} \end{bmatrix} = (I + \Delta t A) \begin{bmatrix} v^k \\ \gamma^k \end{bmatrix} + \Delta t B$$

Método implícito

$$\begin{bmatrix} \frac{v^{k+1}-v^k}{\Delta t} \\ \frac{\gamma^{k+1}-\gamma^k}{\Delta t} \end{bmatrix} = \begin{bmatrix} \frac{-\rho \cdot v_0 \cdot S \cdot CD}{m} & -g \\ \frac{0.5 \cdot \rho \cdot S \cdot v_0^2 + m \cdot g}{m \cdot v_0^2} & 0 \end{bmatrix} \begin{bmatrix} v^{k+1} \\ \gamma^{k+1} \end{bmatrix} + \begin{bmatrix} \frac{\rho \cdot v_0 \cdot S \cdot CD}{m} \\ -2\frac{g}{v_0} \end{bmatrix}$$

$$\begin{bmatrix} v^{k+1} \\ \gamma^{k+1} \end{bmatrix} = (I - \Delta t A)^{-1} \begin{bmatrix} v^k \\ \gamma^k \end{bmatrix} + (I - \Delta t A)^{-1} B$$

Método Crank-Nicolson

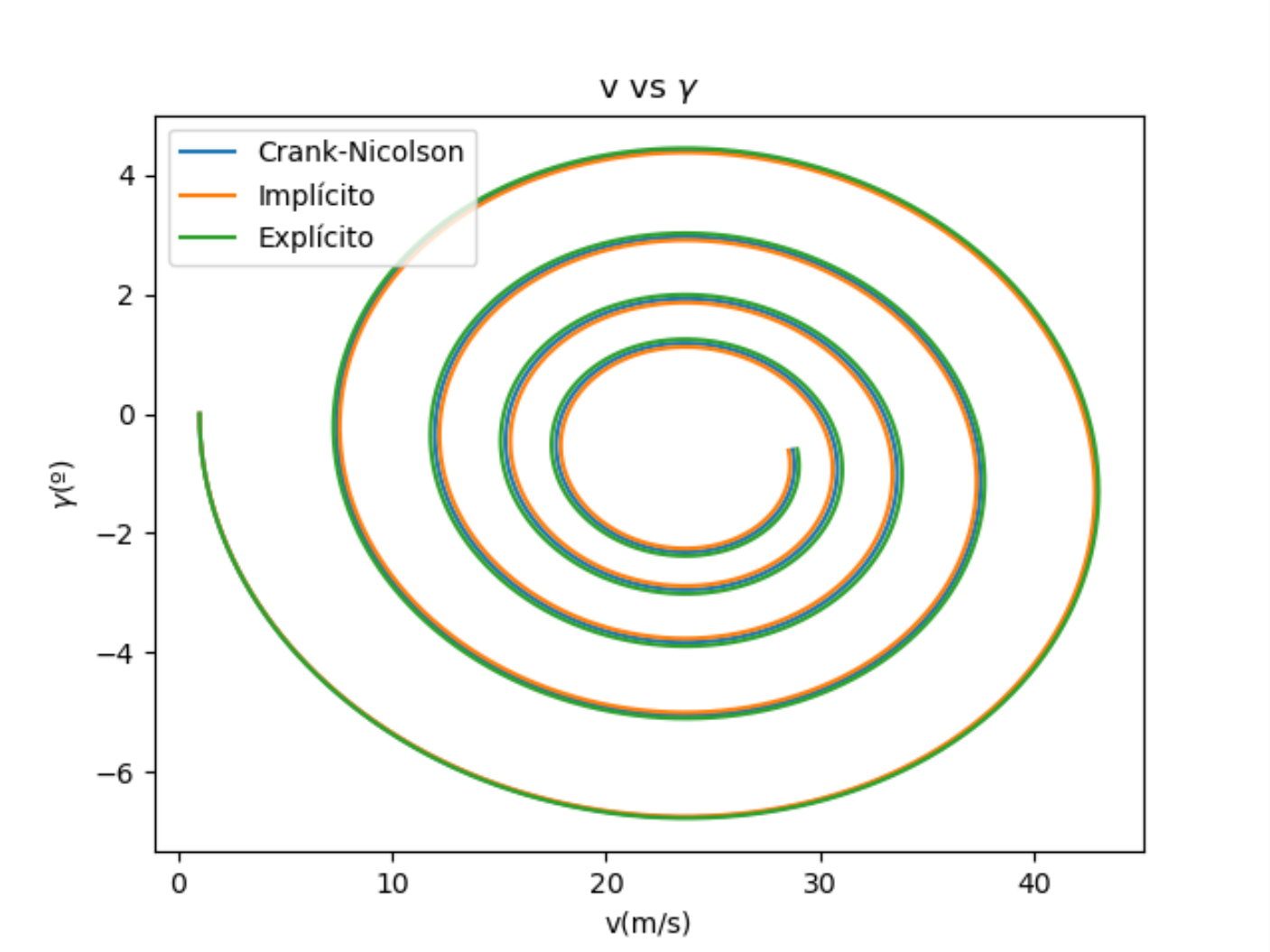
$$\begin{bmatrix} \frac{v^{k+1}-v^k}{\Delta t} \\ \frac{\gamma^{k+1}-\gamma^k}{\Delta t} \end{bmatrix} = \begin{bmatrix} \frac{-\rho \cdot v_0 \cdot S \cdot CD}{m} & -g \\ \frac{0.5 \cdot \rho \cdot S \cdot v_0^2 + m \cdot g}{m \cdot v_0^2} & 0 \end{bmatrix} \begin{bmatrix} \frac{v^{k+1}+v^k}{2} \\ \frac{\gamma^{k+1}+\gamma^k}{2} \end{bmatrix} + \begin{bmatrix} \frac{\rho \cdot v_0 \cdot S \cdot CD}{m} \\ -2\frac{g}{v_0} \end{bmatrix}$$

$$\begin{bmatrix} v^{k+1} \\ \gamma^{k+1} \end{bmatrix} = \left(I - \frac{\Delta t A}{2}\right)^{-1} \left(I + \frac{\Delta t A}{2}\right) \begin{bmatrix} v^k \\ \gamma^k \end{bmatrix} + \Delta t \left(I - \frac{\Delta t A}{2}\right)^{-1} B$$

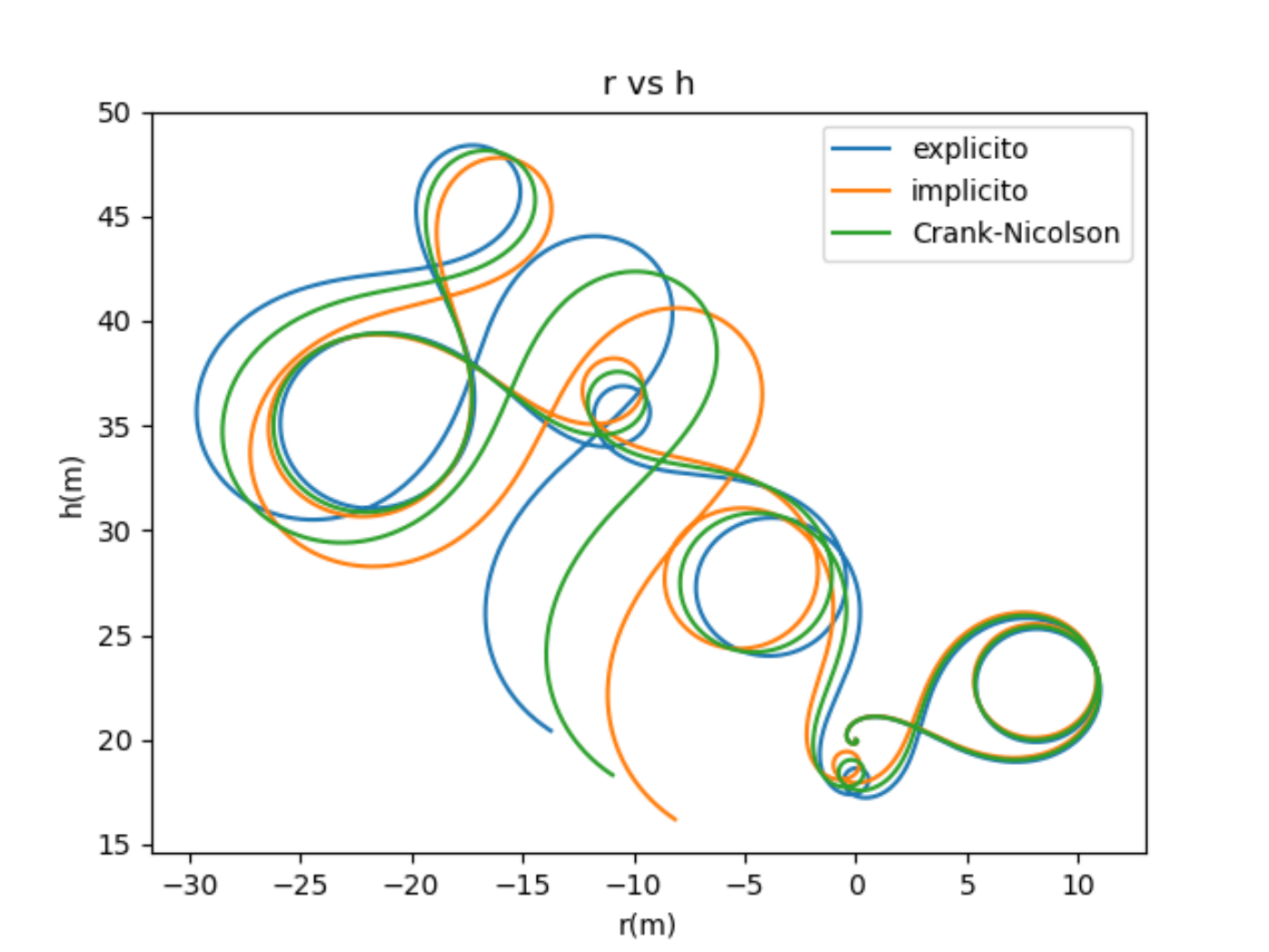
SOLUCIÓN CON MÉTODOS:

EXPLÍCITO, IMPLÍCITO Y CRANK-NICOLSON

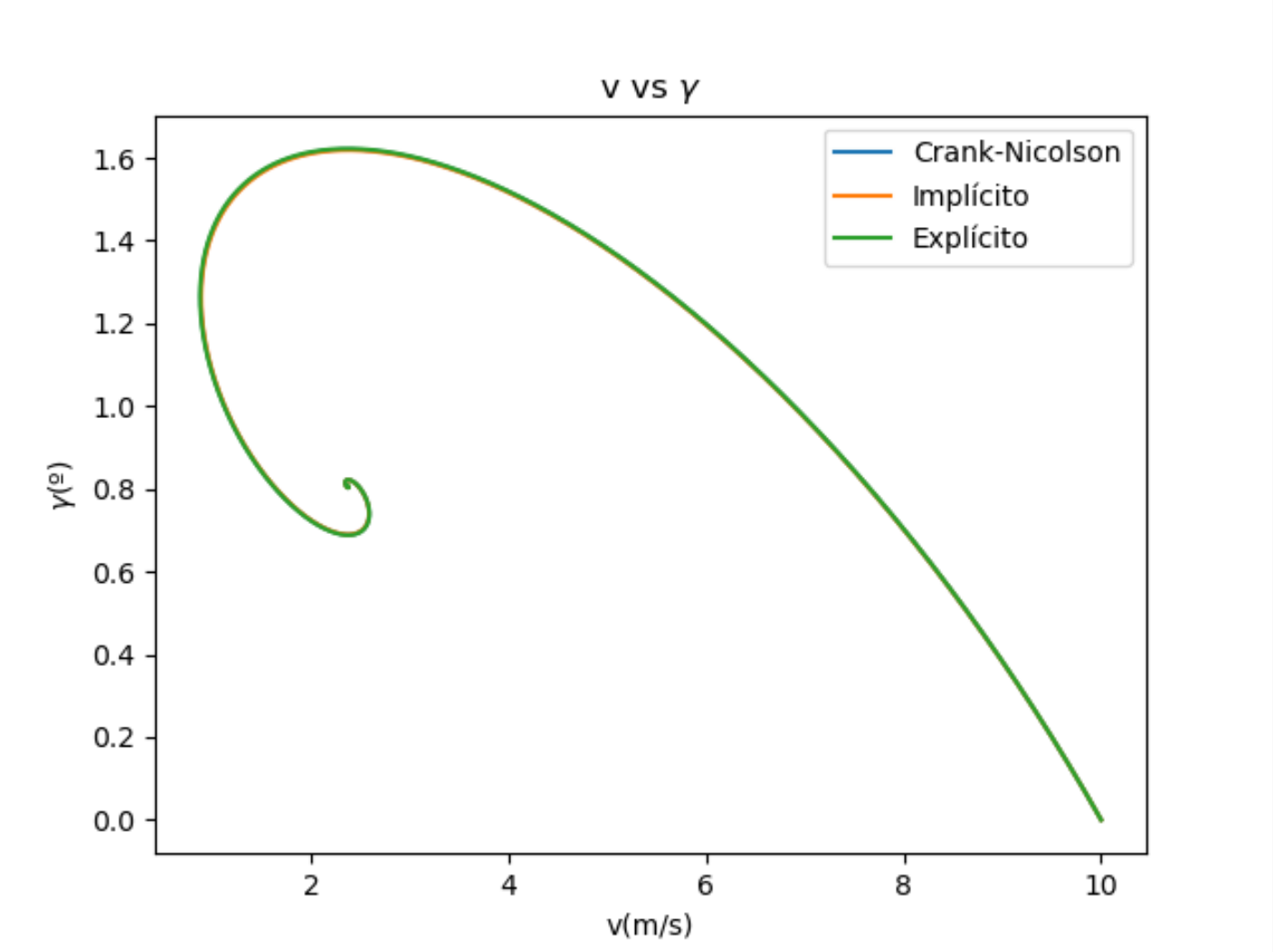
Ángulo frente a velocidad, $\boldsymbol{v} = 1$ m/s



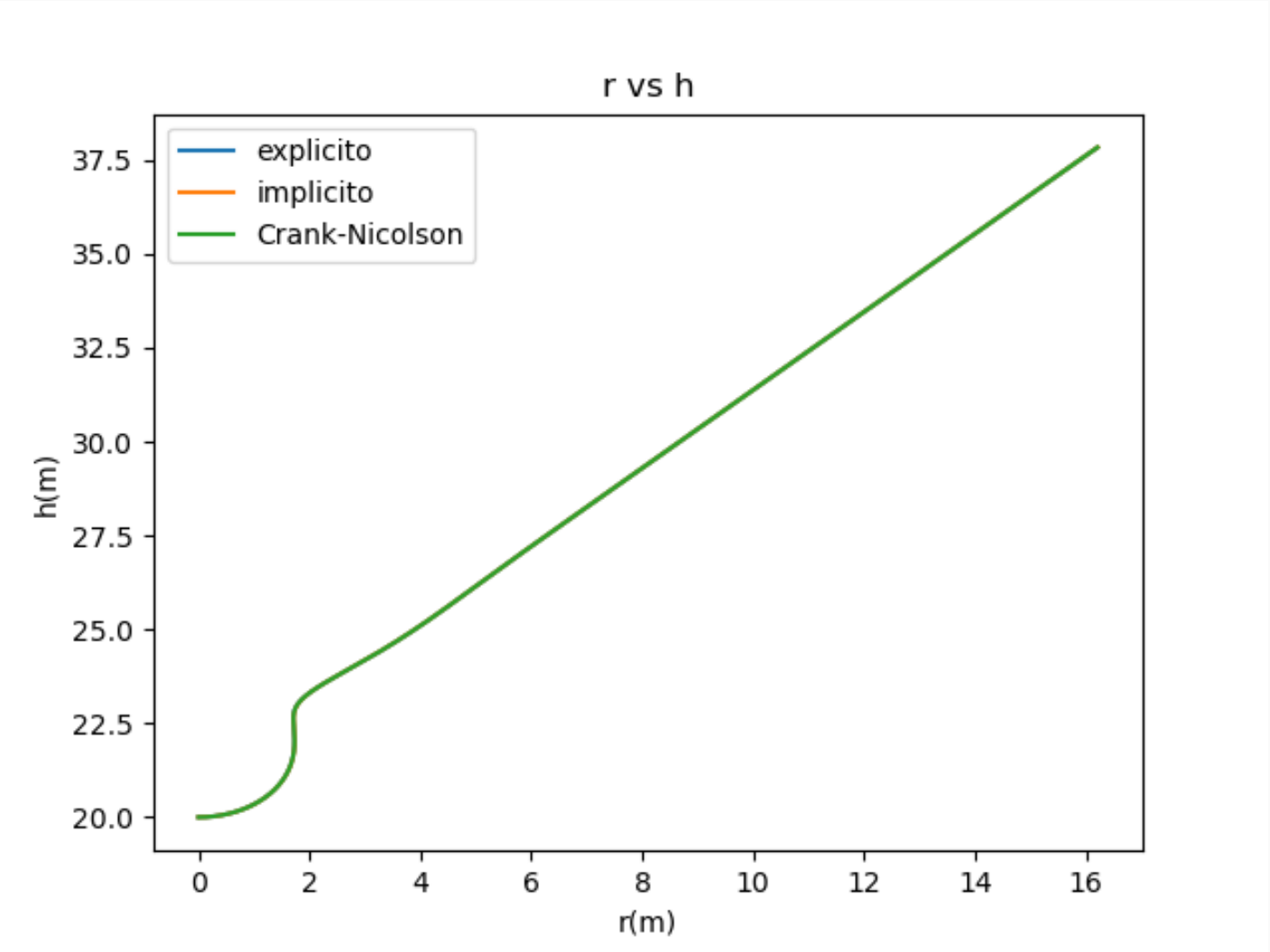
Trayectoria, $\boldsymbol{v} = 1$ m/s



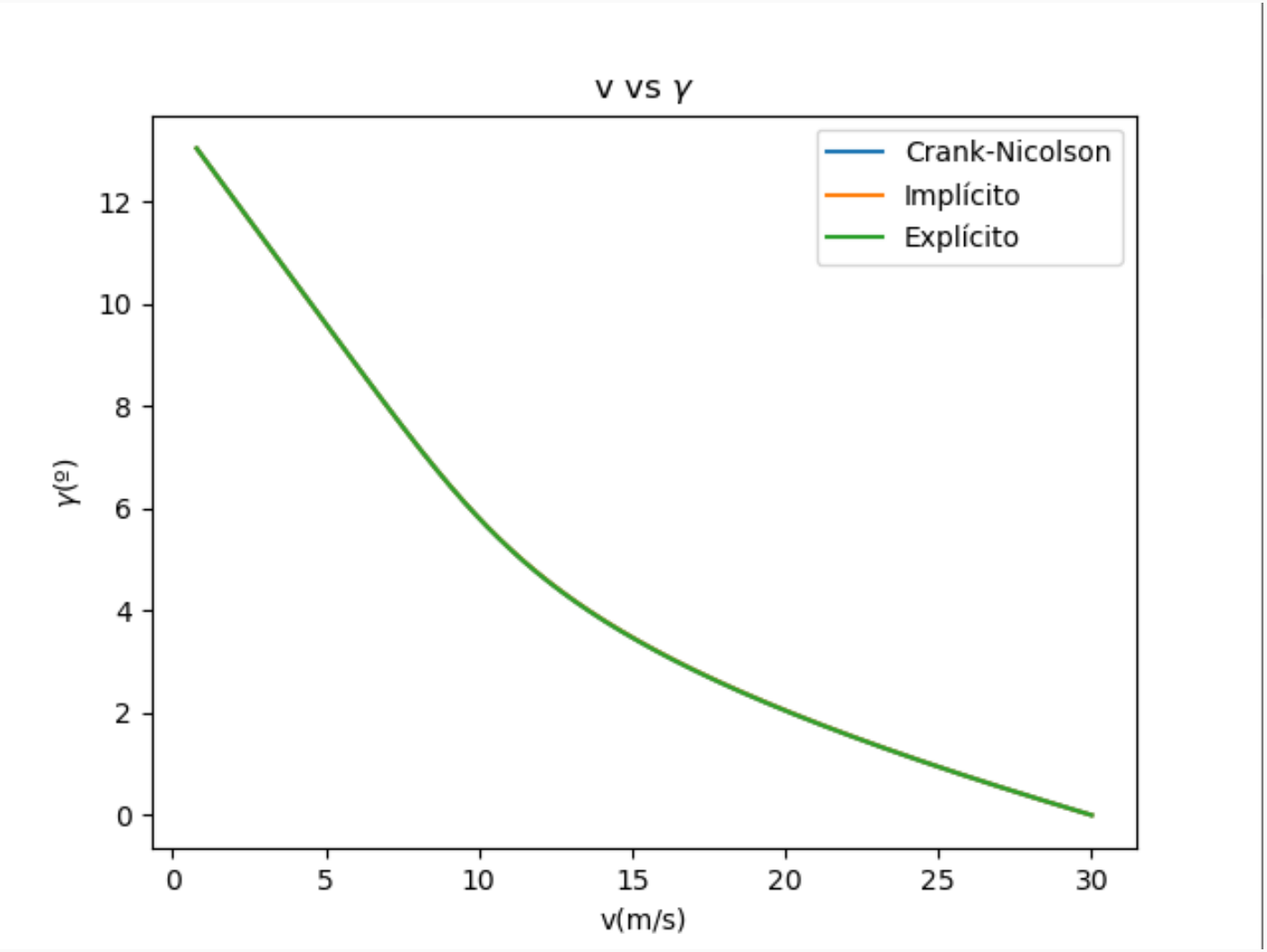
Ángulo frente a velocidad, $\boldsymbol{v} = 10$ m/s



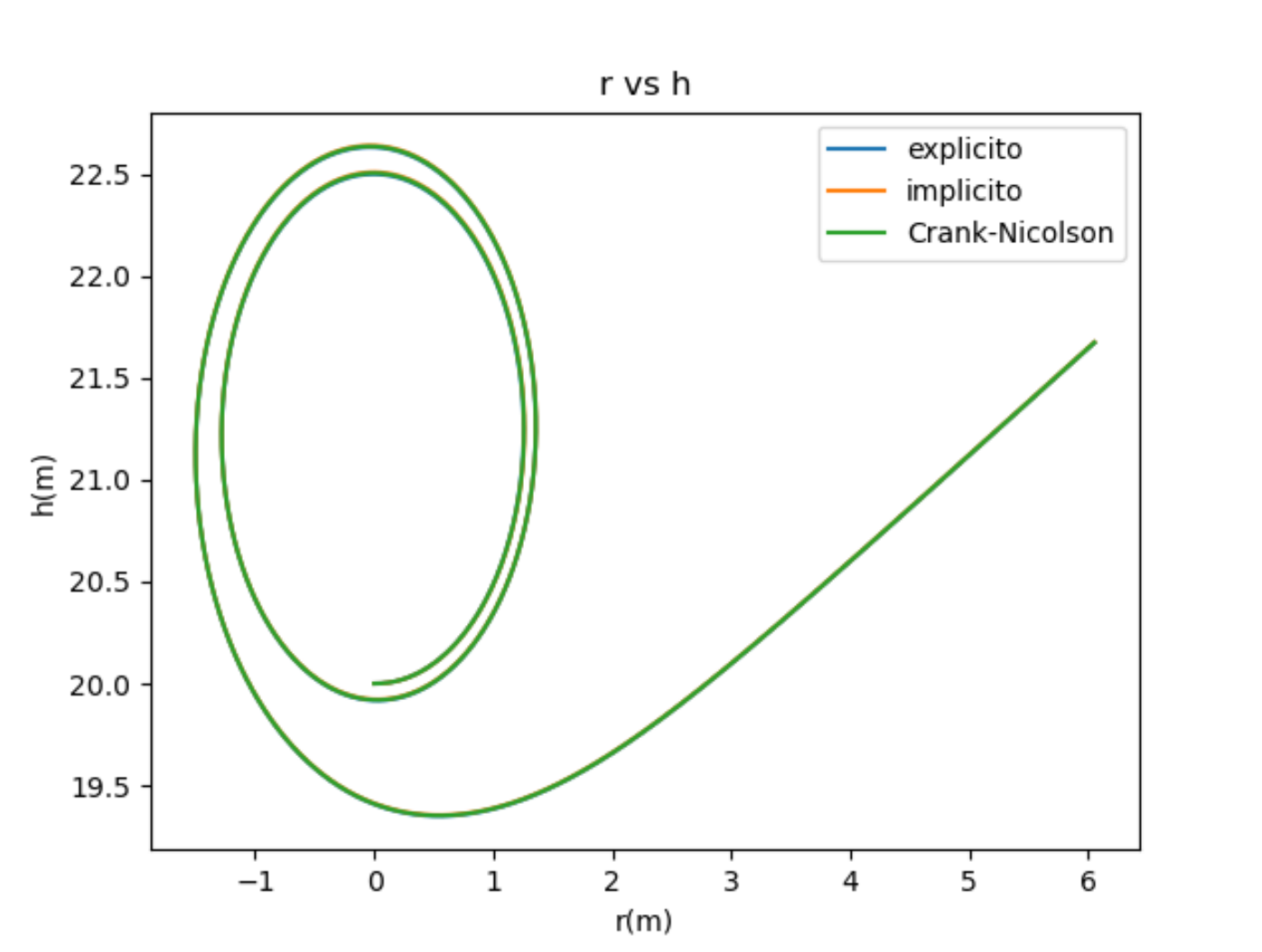
Trayectoria, $\boldsymbol{v} = 10$ m/s



Ángulo frente a velocidad, $\boldsymbol{v} = 30$ m/s

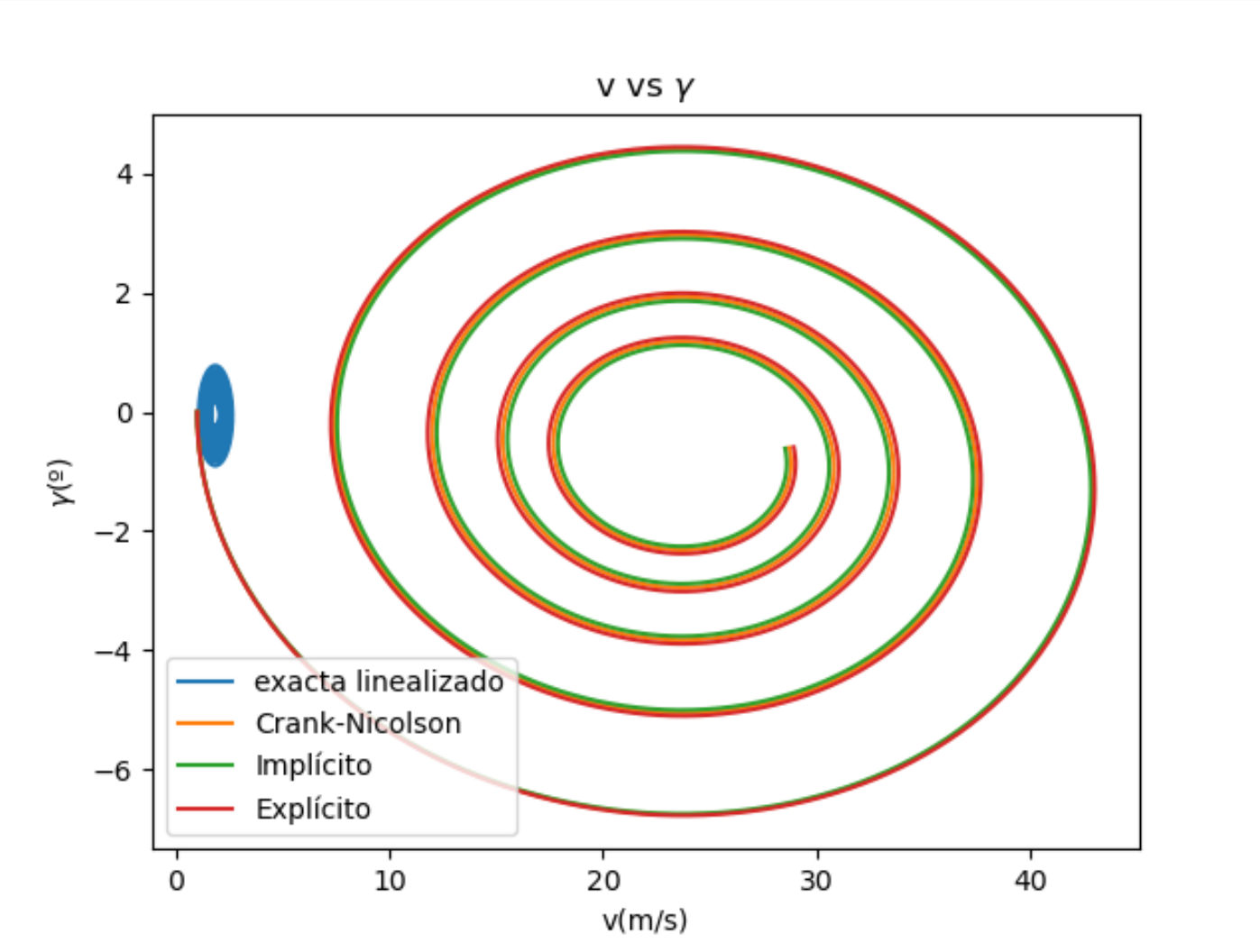


Trayectoria, $\boldsymbol{v} = 30$ m/s

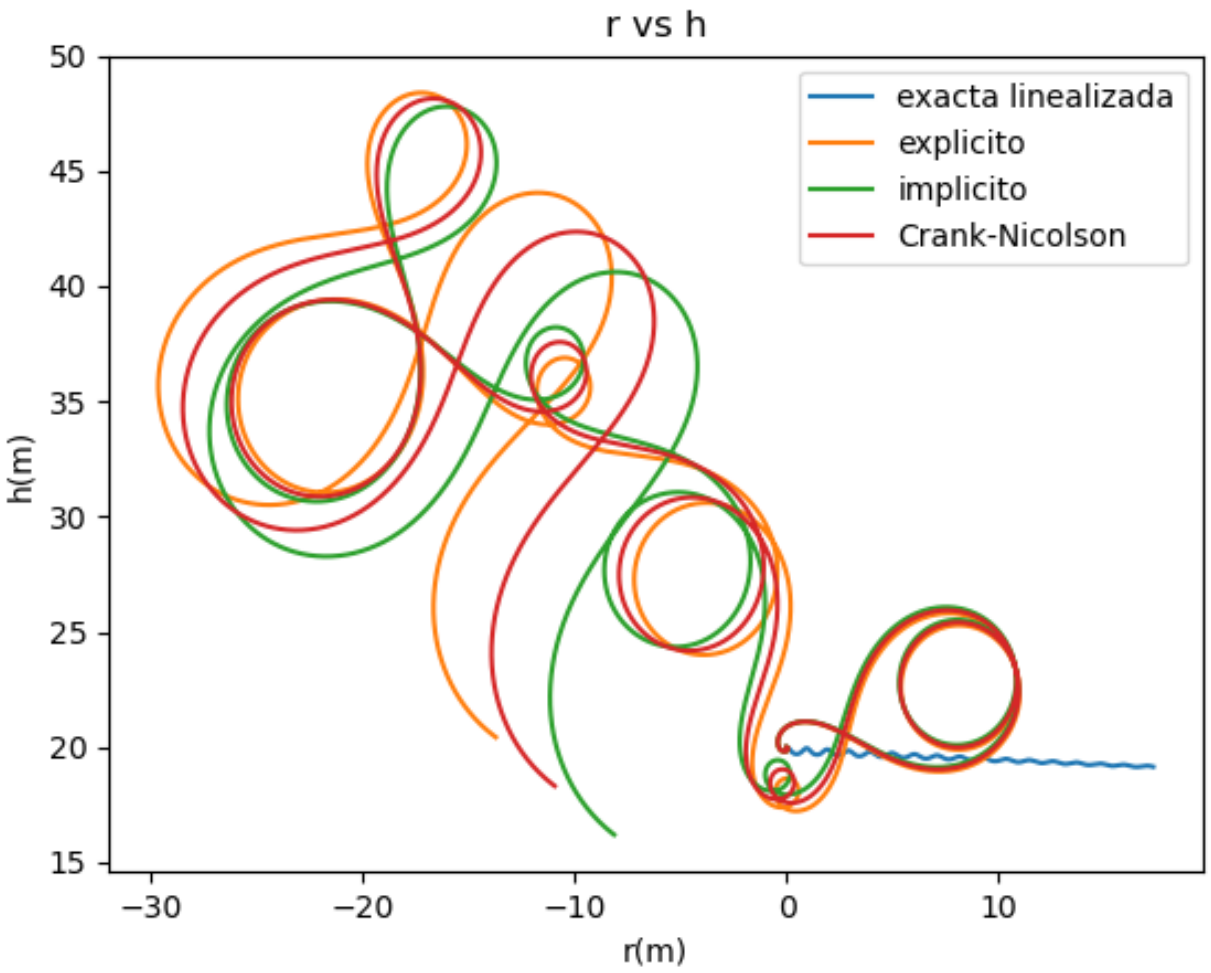


COMPARACIÓN CON ODEINT

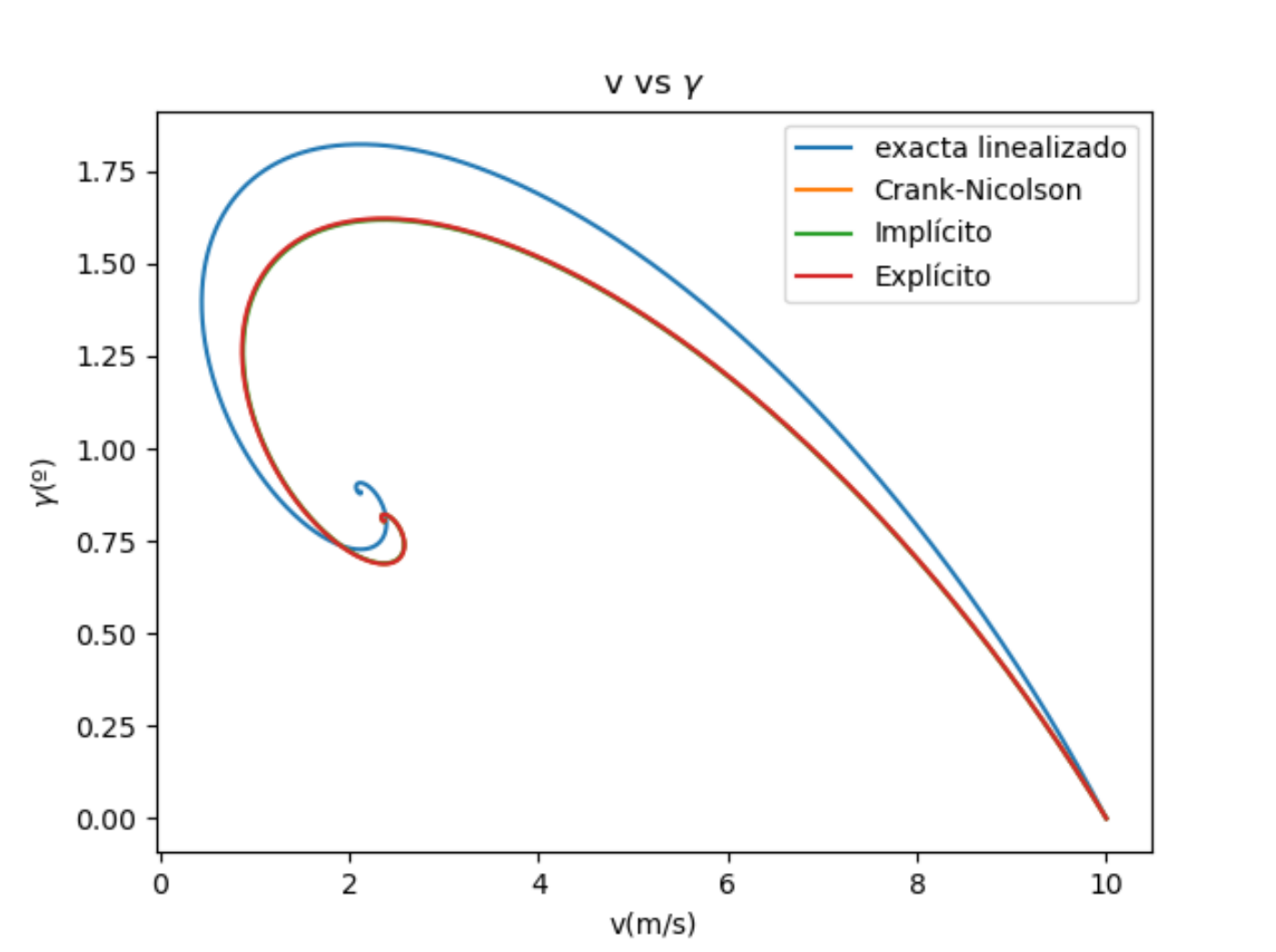
Ángulo frente a velocidad, $\boldsymbol{v} = 1$ m/s



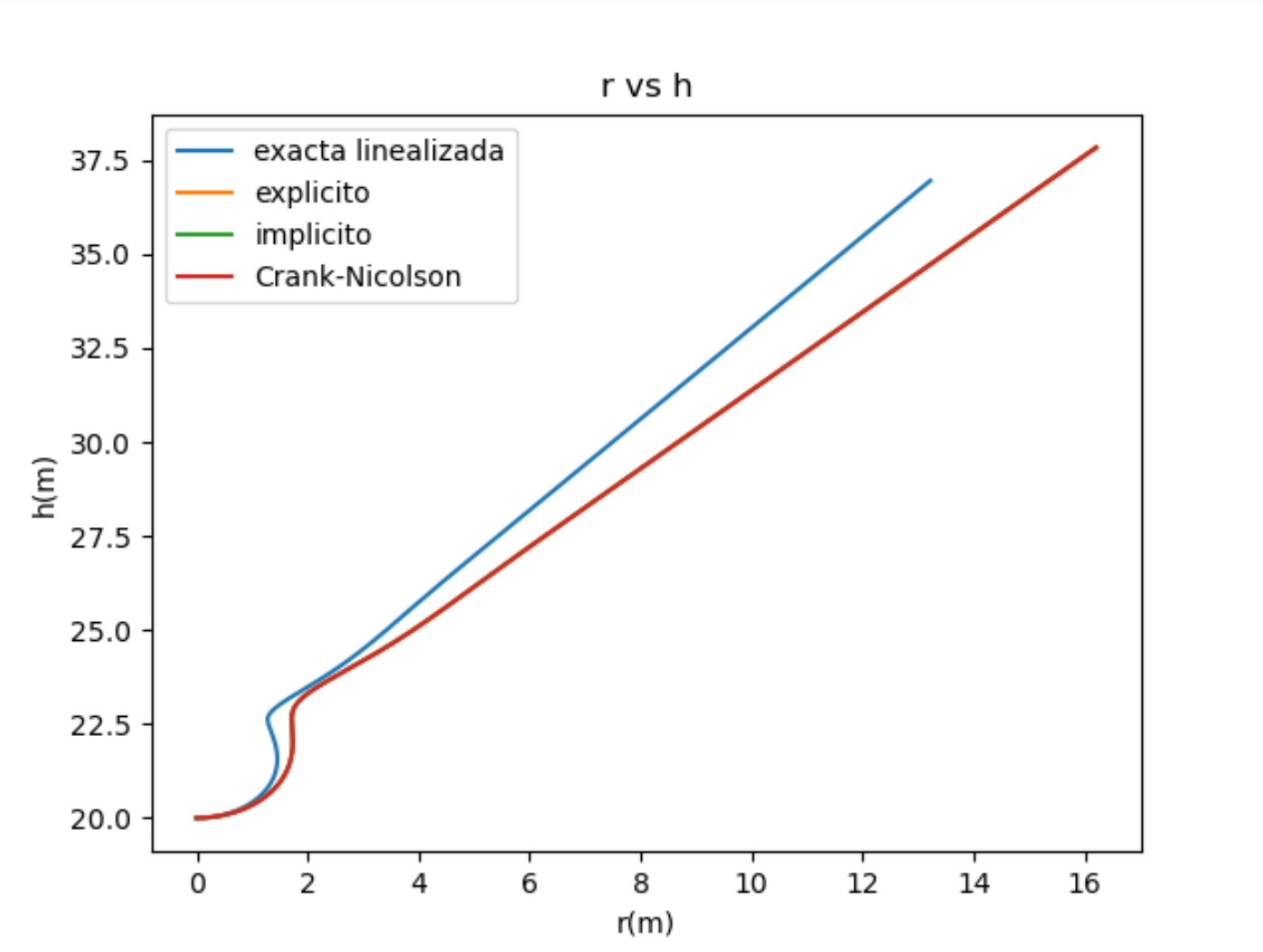
Trayectoria, $\boldsymbol{v} = 1$ m/s



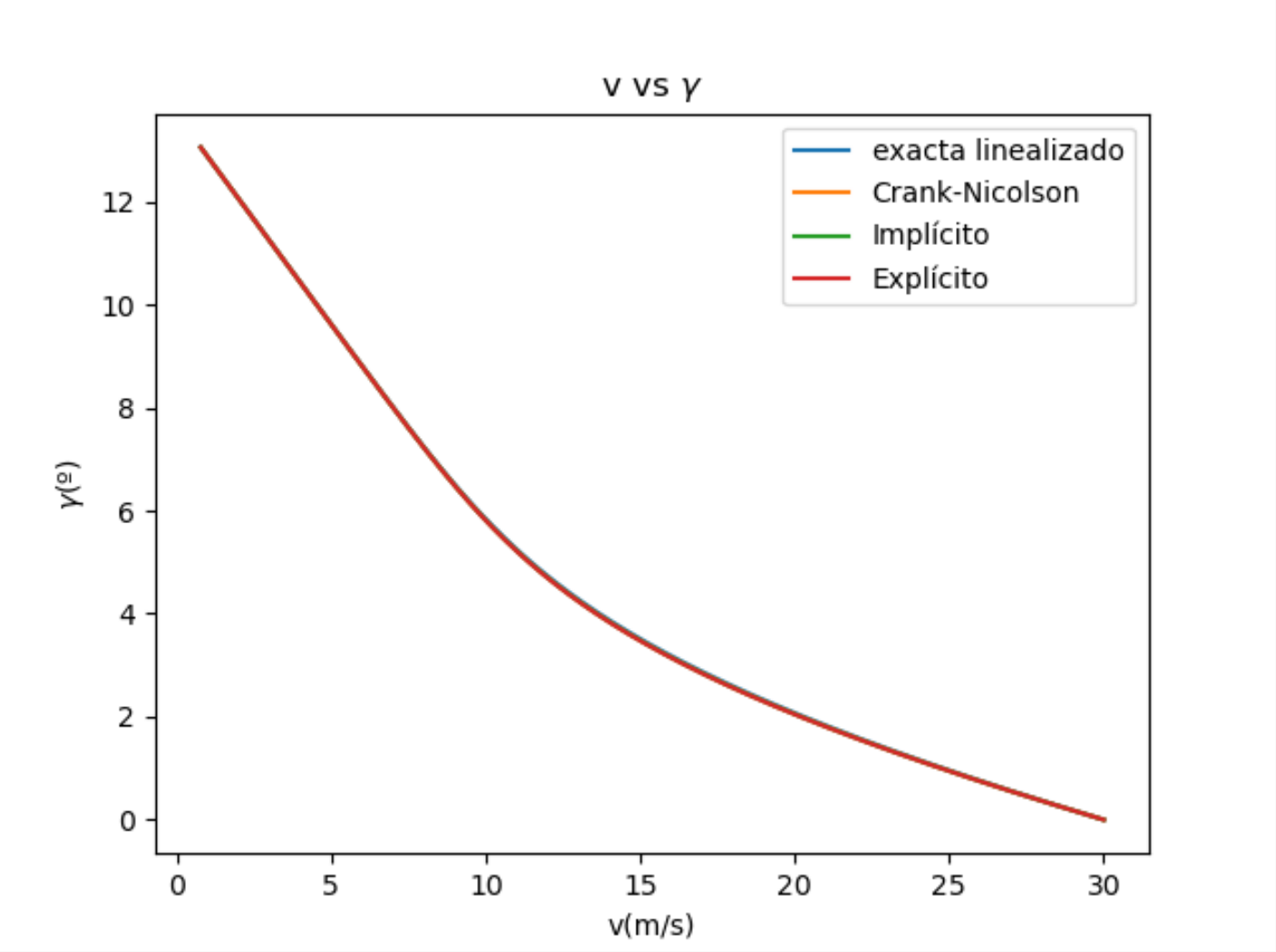
Ángulo frente a velocidad, $\boldsymbol{v} = 10$ m/s



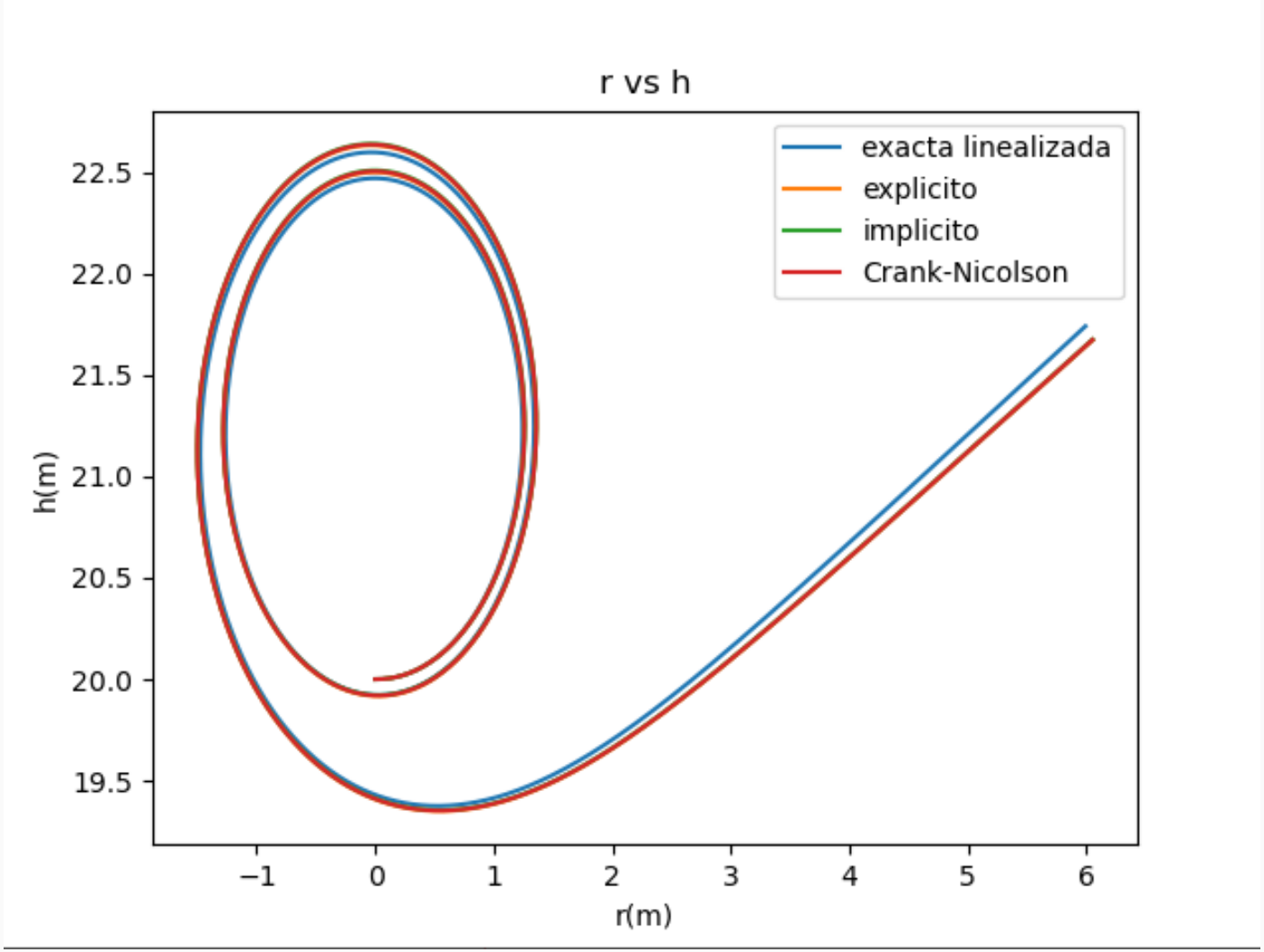
Trayectoria, $\boldsymbol{v} = 10$ m/s



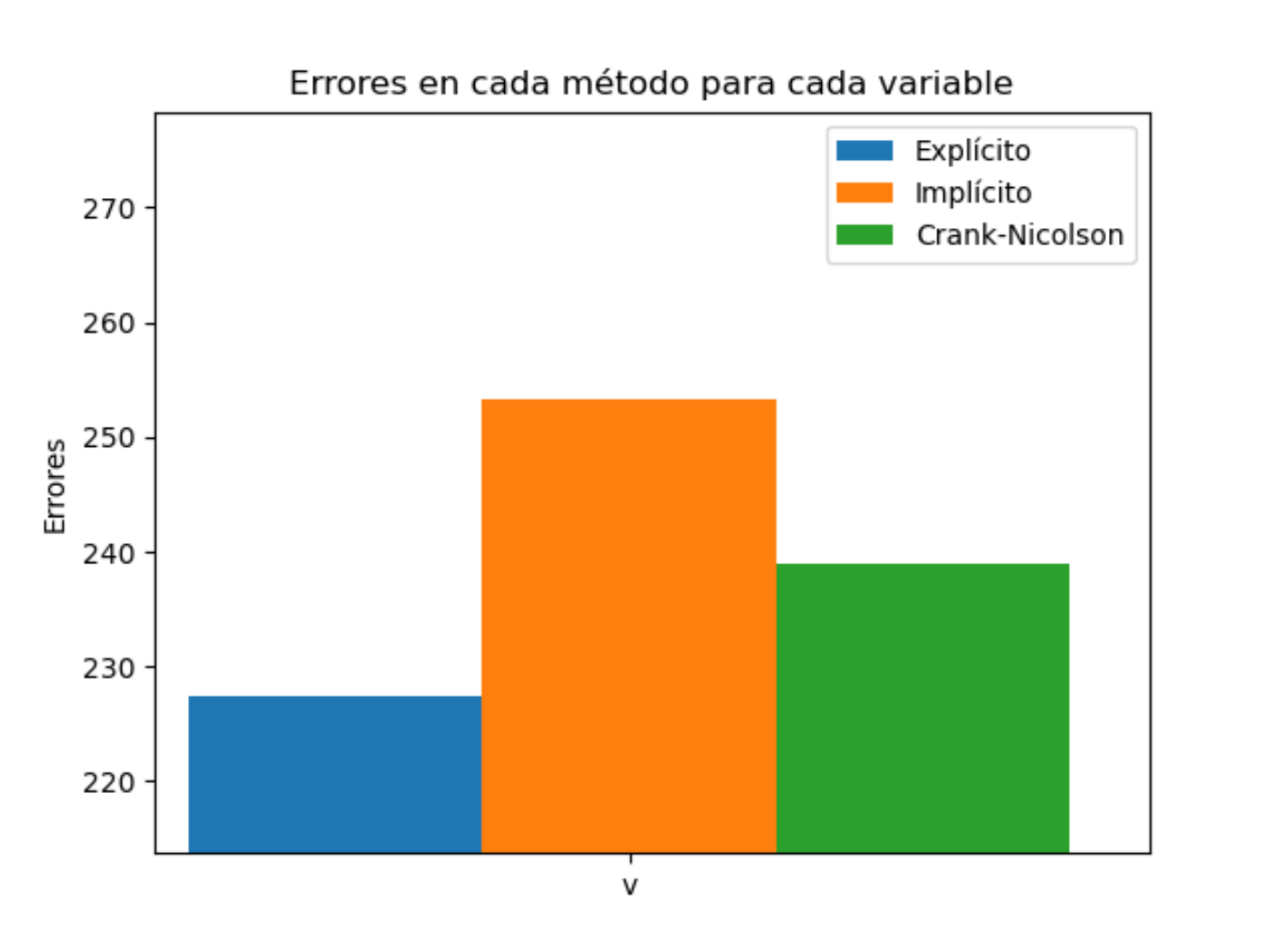
Ángulo frente a velocidad, $\boldsymbol{v} = 30$ m/s



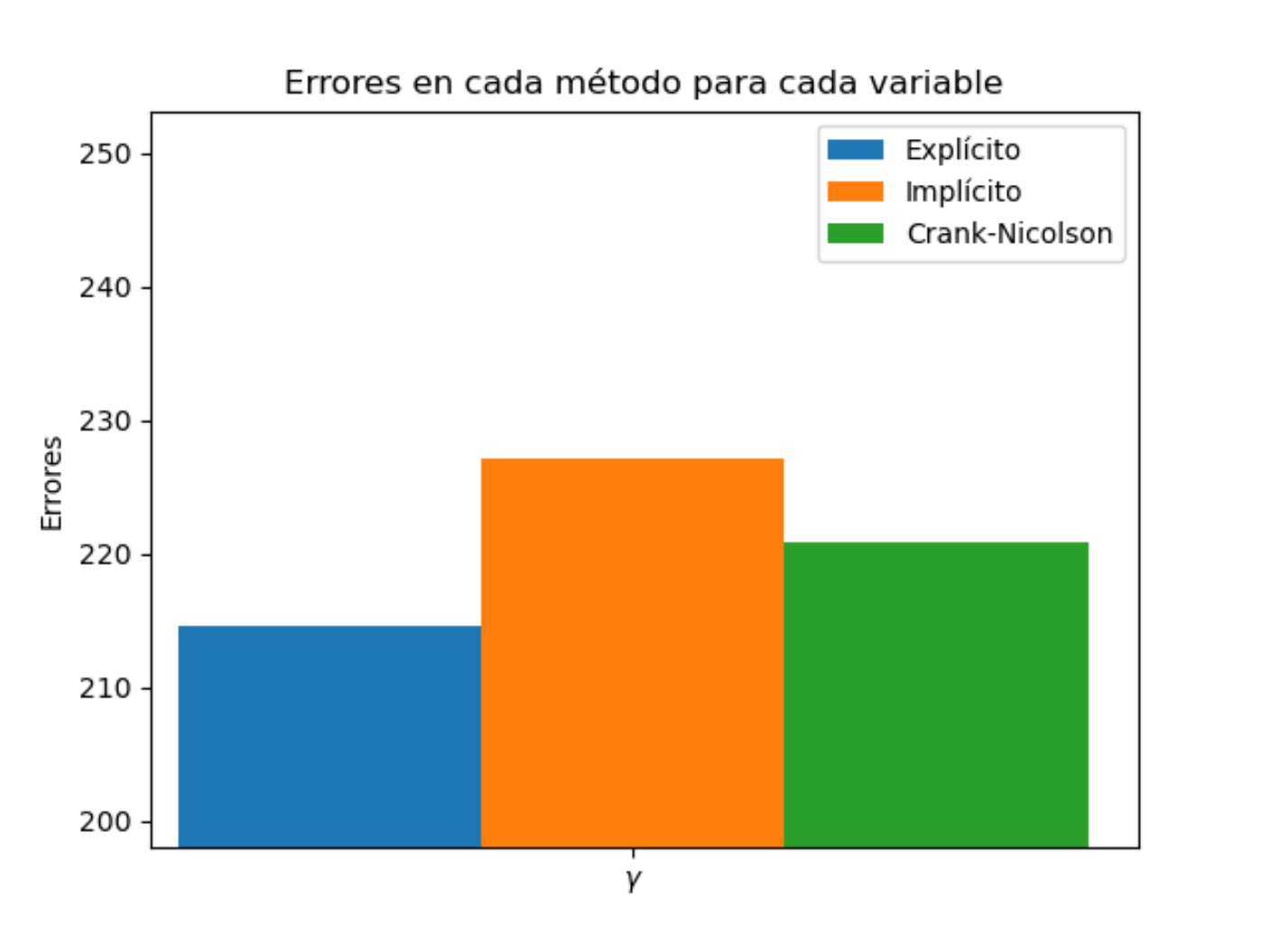
Trayectoria, $\boldsymbol{v} = 30$ m/s



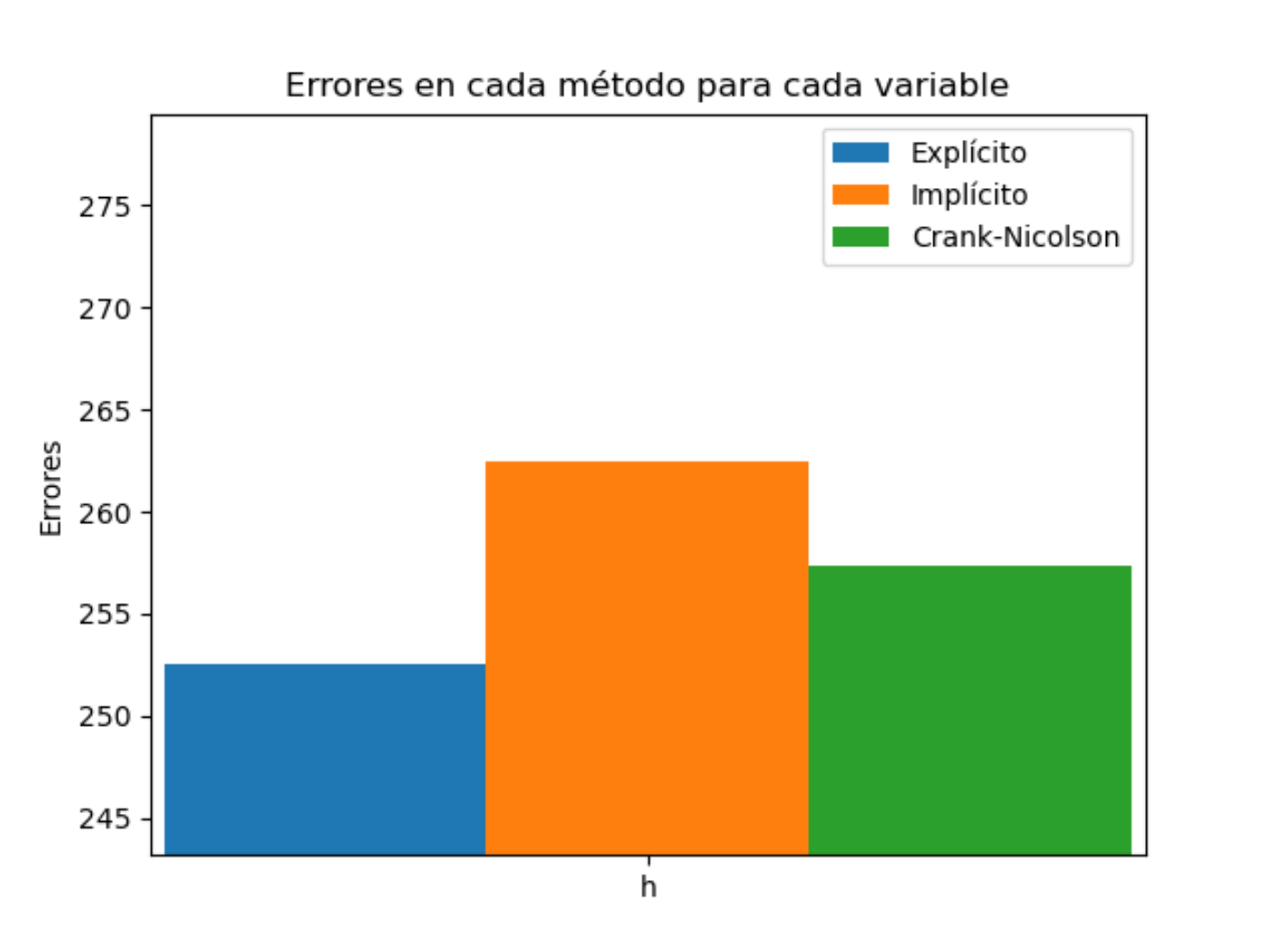
Error total a tiempo corto de la velocidad



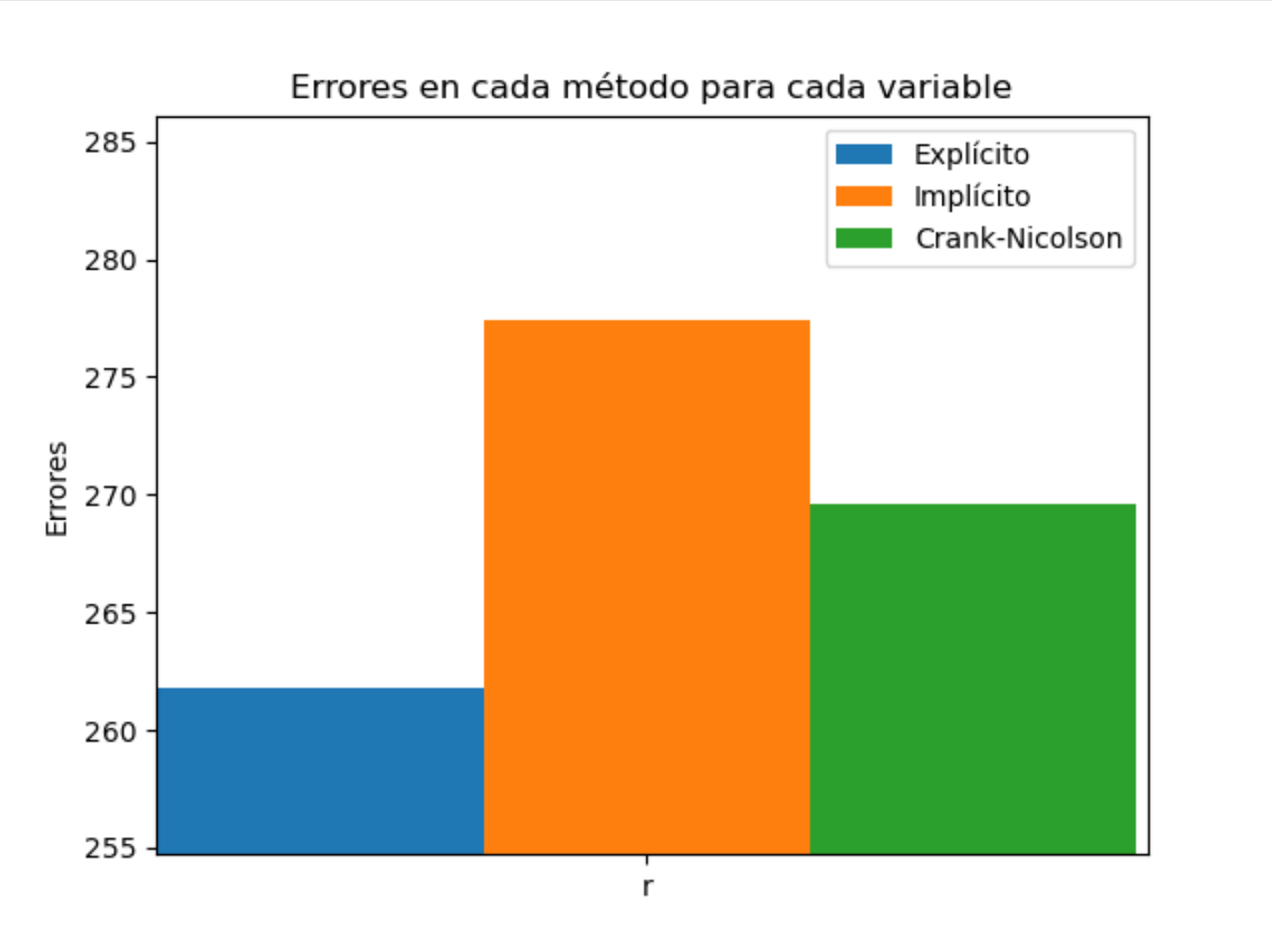
Error total a tiempo corto de la gamma



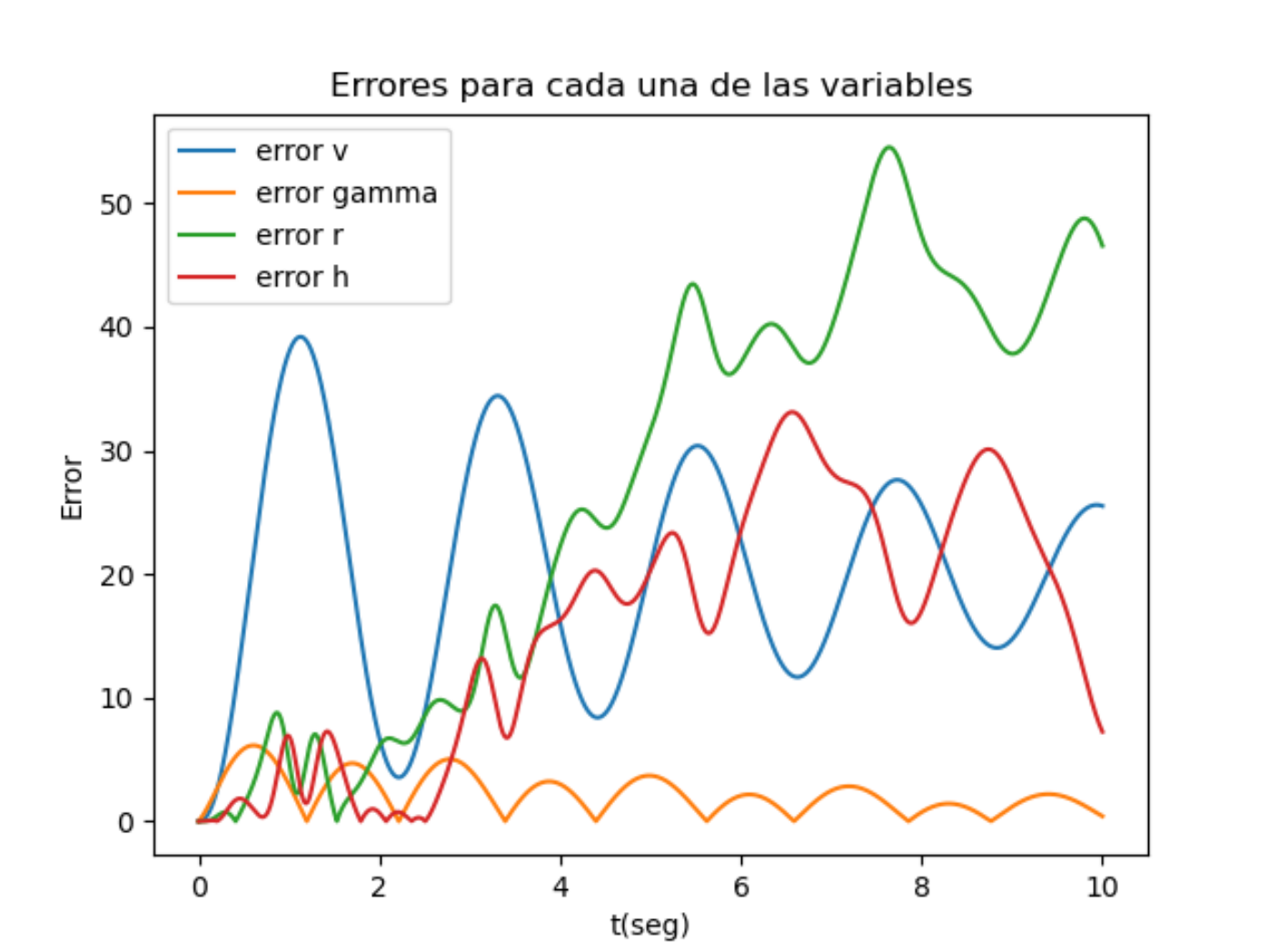
Error total a tiempo corto de la h



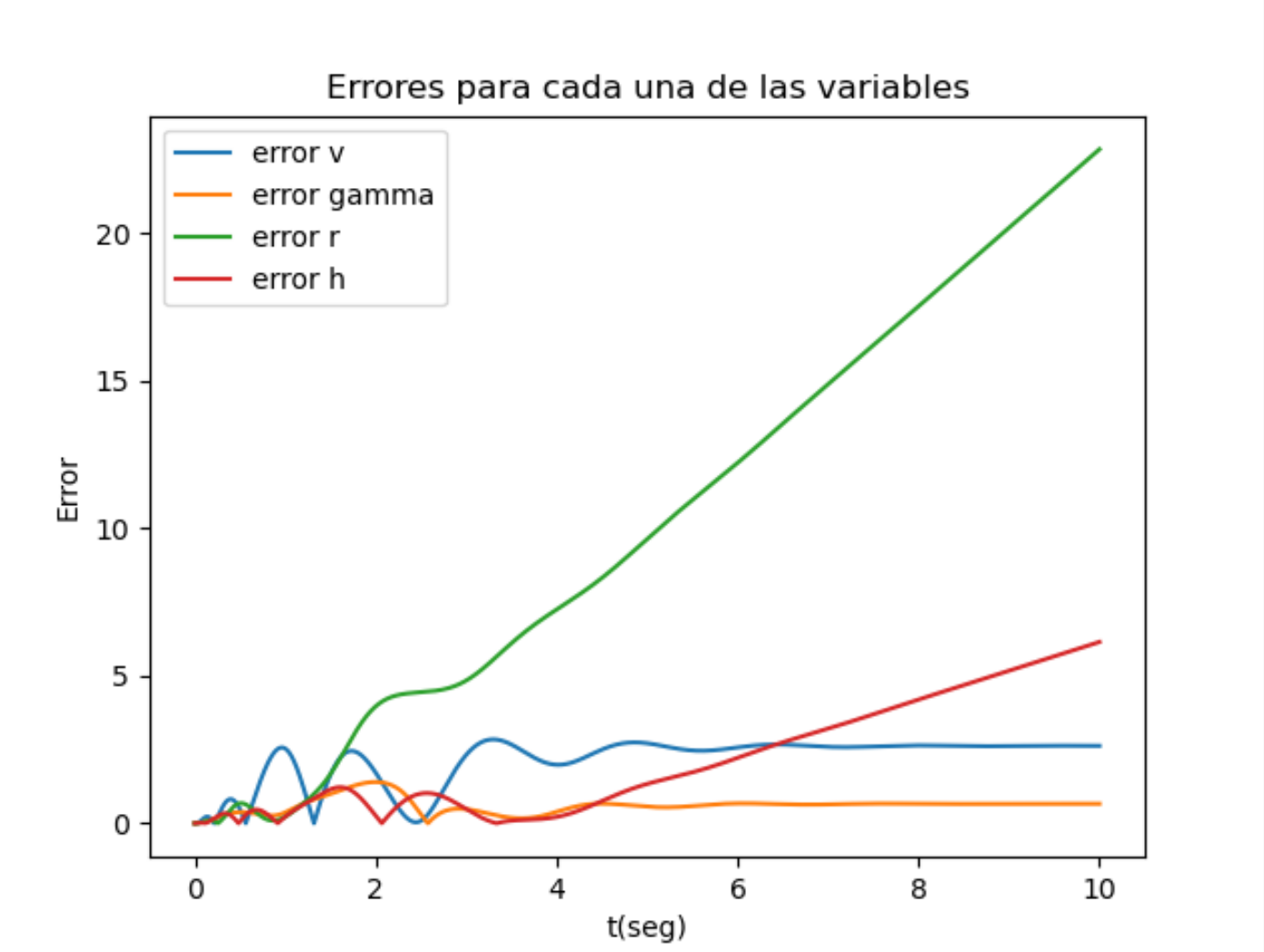
Error total a tiempo corto de la r



Errores explícito - odeint $\boldsymbol{v}_0 = 1$ m/s

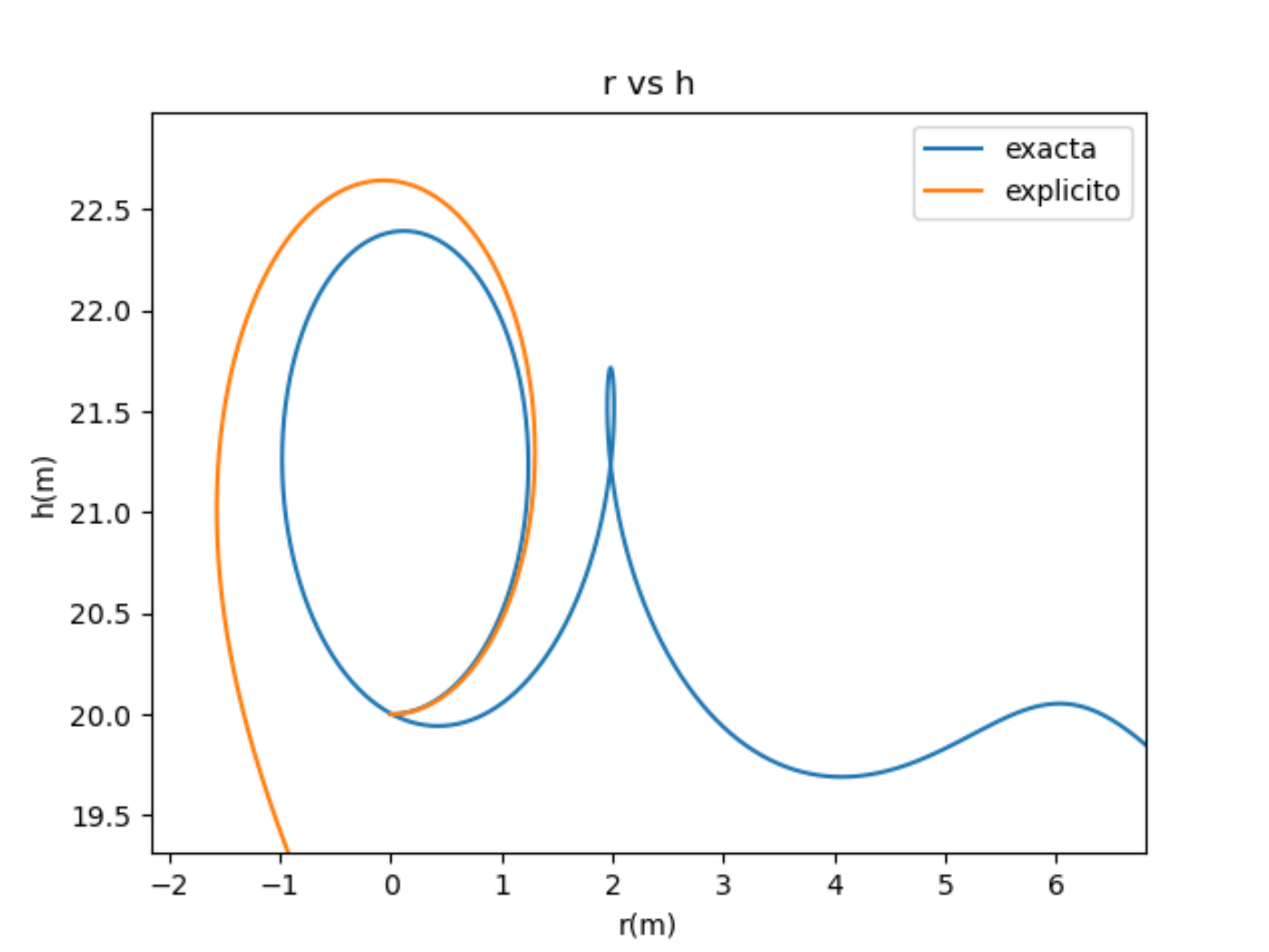


Errores explícito - odeint $v_0 = 30$ m/s

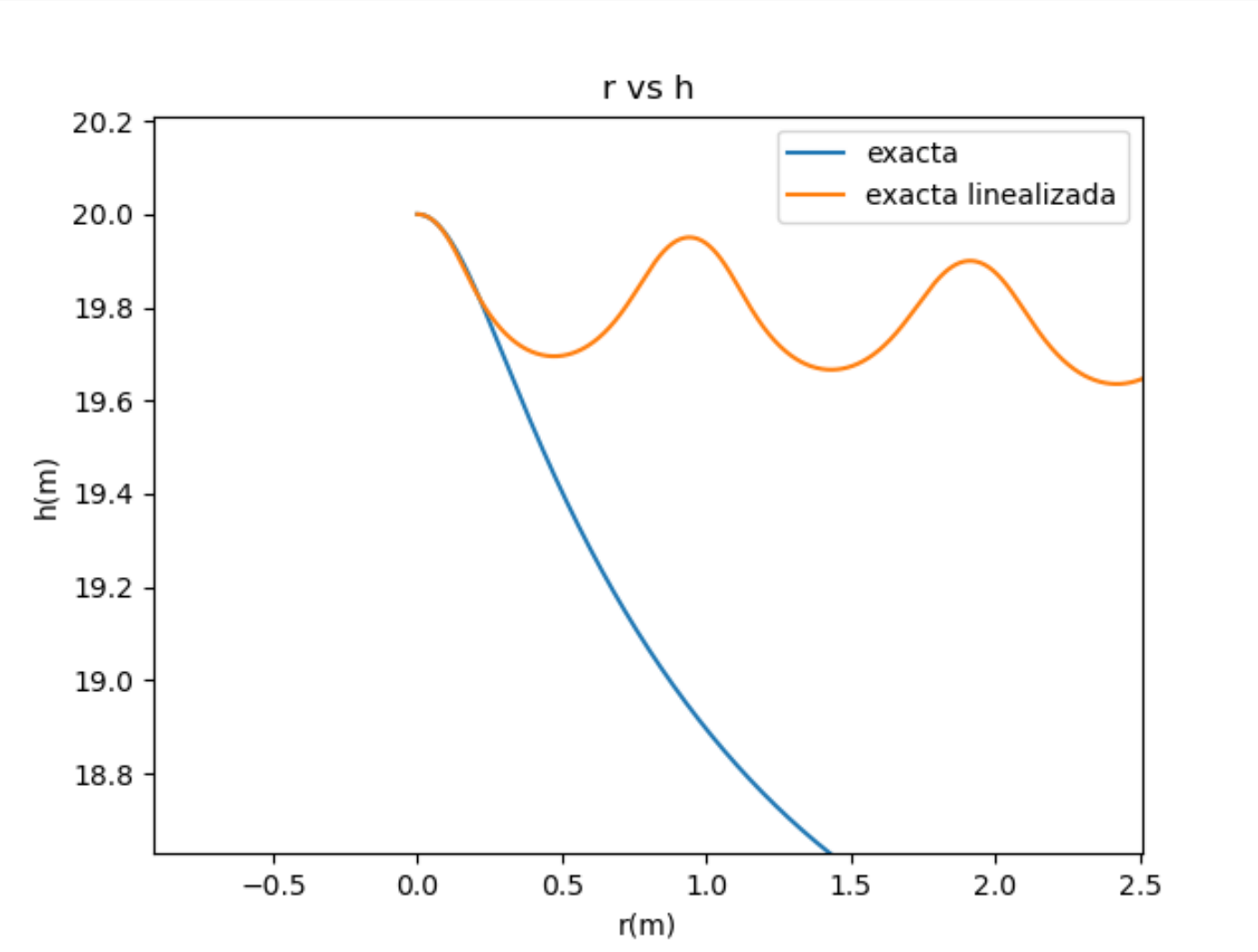


CONCLUSIÓN

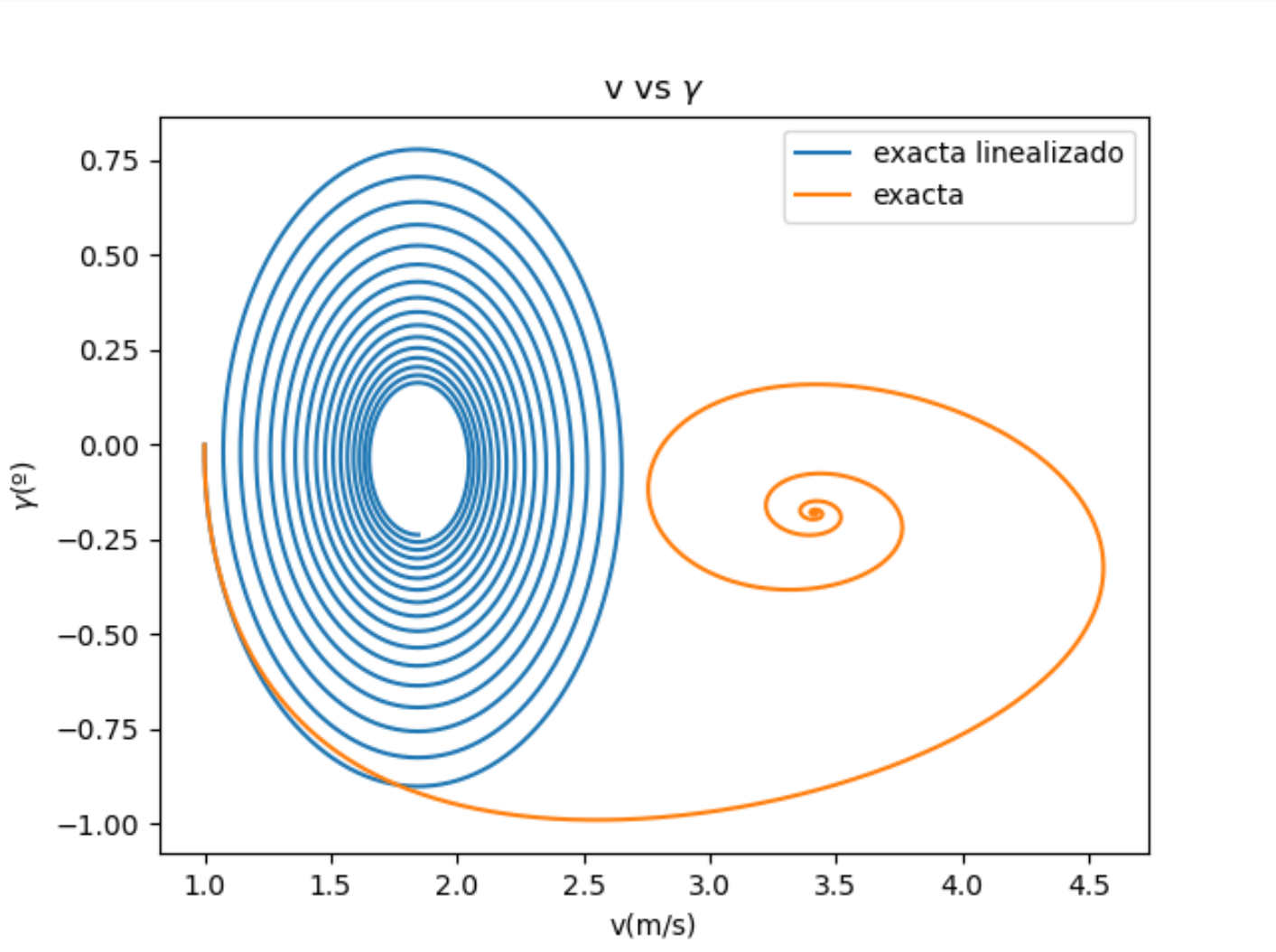
Trayectoria $\boldsymbol{v}_0 = 20$ m/s



Trayectoria $\boldsymbol{v}_0 = 1$ m/s exactas



v vs γ $\boldsymbol{v}_0 = 1$ m/s exactas

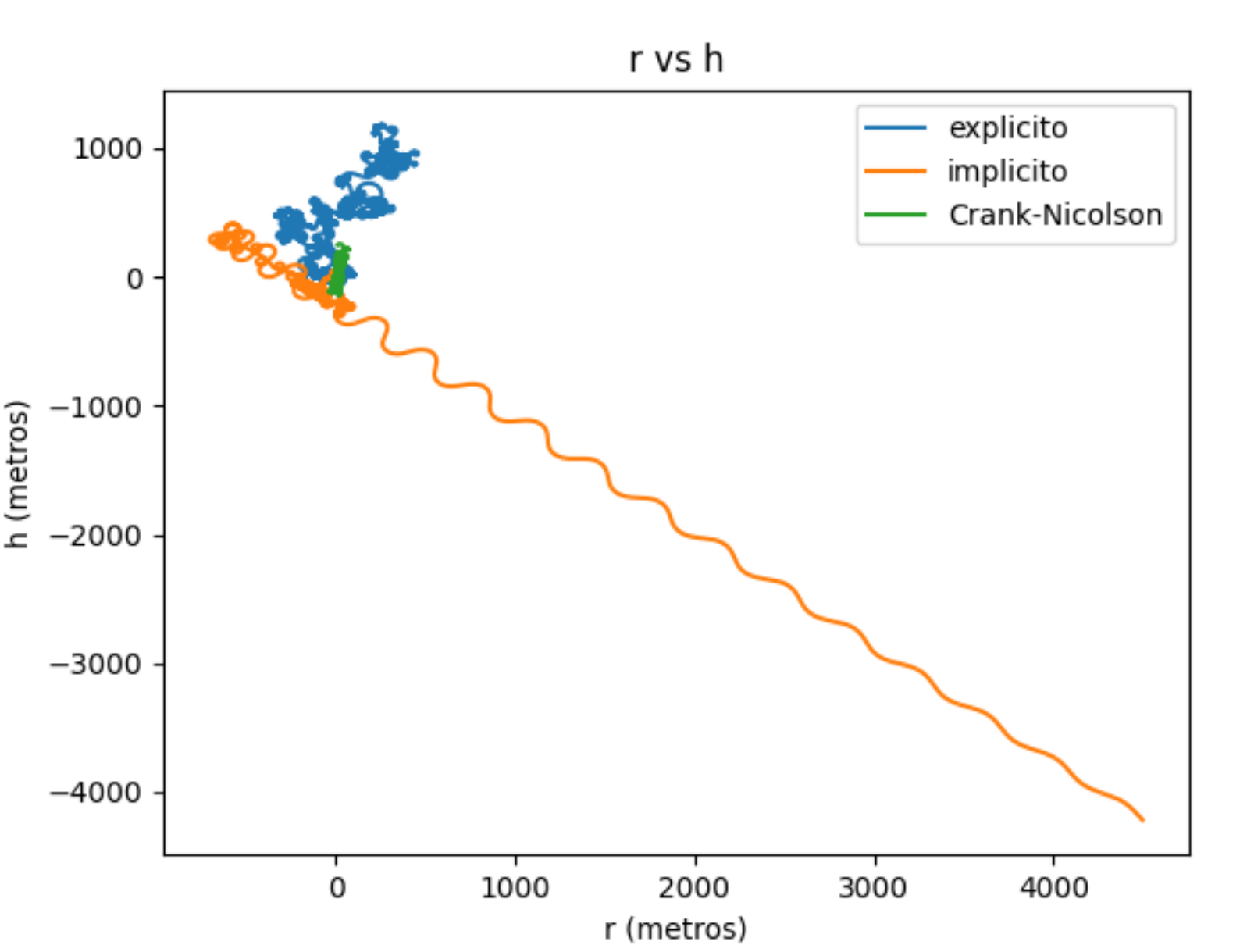


REFERENCIAS

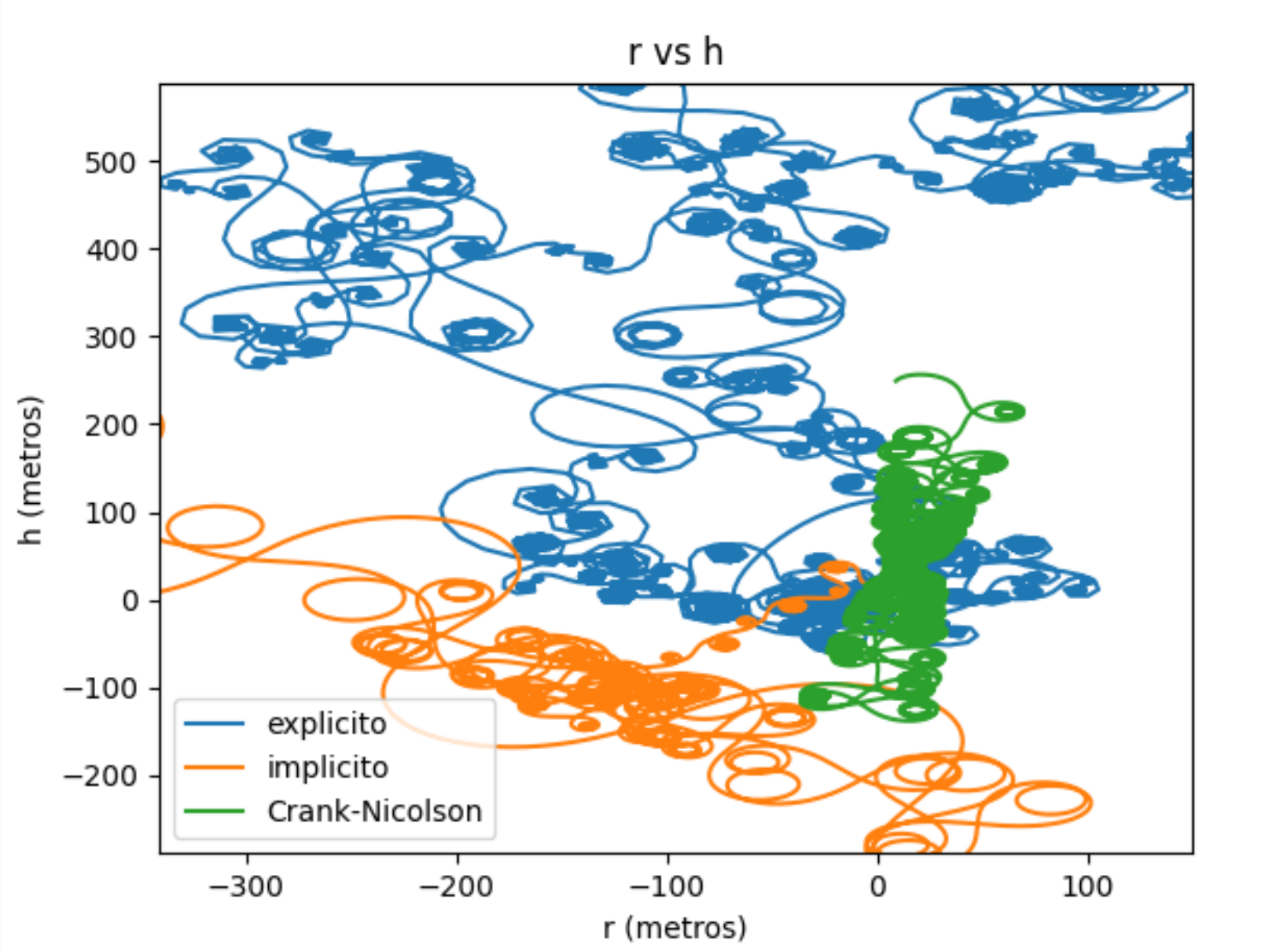
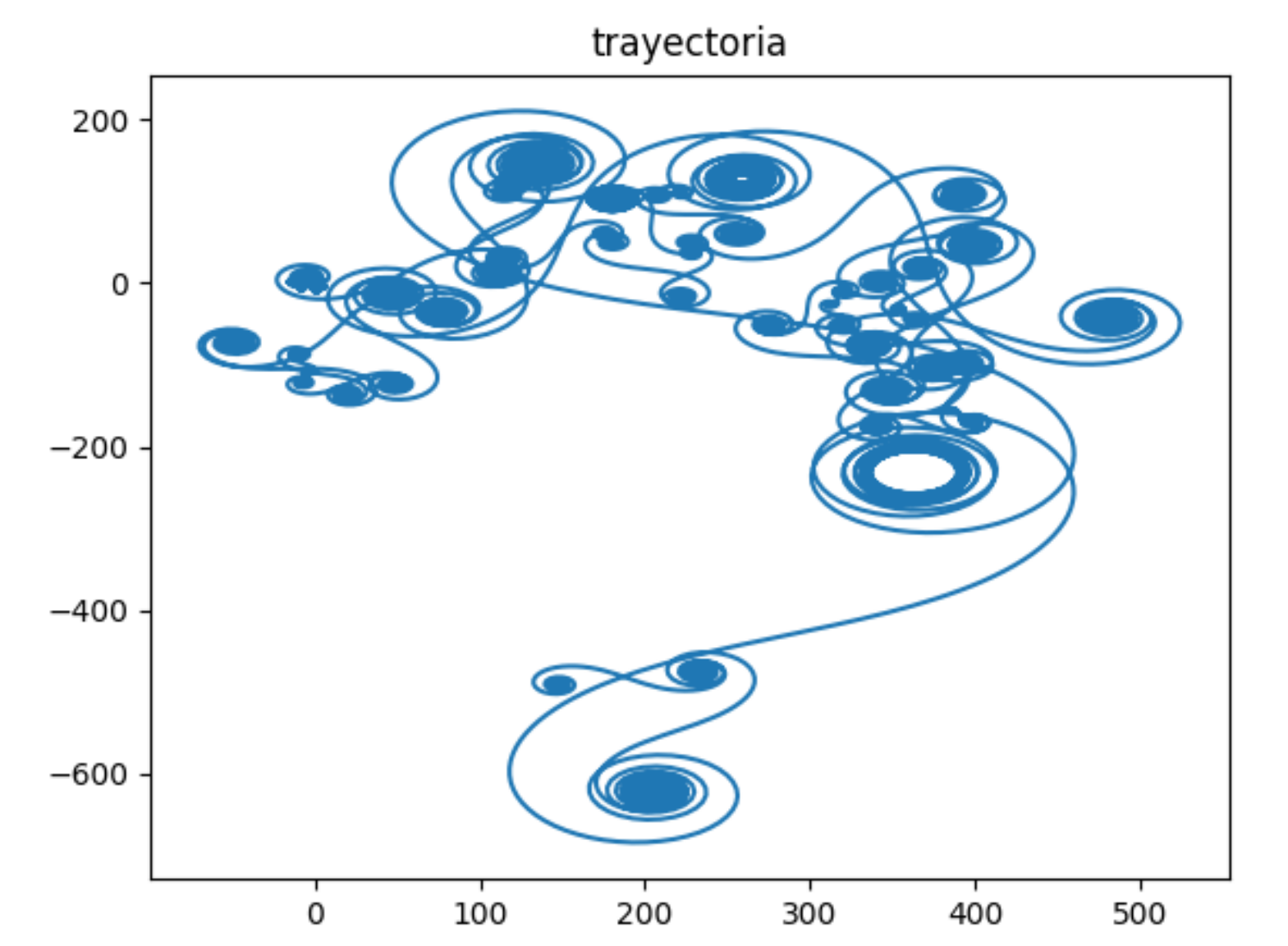
- Apuntes de la asignatura MECÁNICA DE VUELO II del quinto curso del grado en Ingeniería Aeroespacial, impartida en la Universitat Politècnica de Catalunya.
- Recurso online (PDF). Ecuación de movimiento y modelo de pequeñas perturbaciones. Capítulo 2.
- Apuntes de la asignatura CÁLCULO NUMÉRICO I del segundo curso del grado en Matemáticas, impartida en la Universidad de Alicante. Curso 2019/2020.
- Apuntes de la asignatura MCEDO del tercer curso del grado en Matemáticas, impartida en la Universidad de Alicante. Curso 2023/2024.

CURIOSIDAD

Trayectoria primer orden, v = 0.1 m/s



Trayectoria primer orden, v = 0.1 m/s



Trayectoria segundo orden, v = 5 m/s